On the equation $n'' = n$ and a problem of Erdős and Barbeau on products of prime–reciprocal sums

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Abstract

Erdős Problem #307 (Barbeau 1976; Erdős–Graham 1980) asks whether there are finite sets P, Q of primes with $\left(\sum_{p\in P}\frac{1}{p}\right)\left(\sum_{q\in Q}\frac{1}{q}\right) = 1$. Writing $a = \prod P$ and $b = \prod Q$, a solution exists exactly when the arithmetic derivative has a two–cycle $a' = b$, $b' = a$ with $a \neq b$. The reformulation is due to Ufnarovski and Åhlander, whose Conjecture 4 states #307 in these terms; the identification of the two problems is recorded in neither literature.

We prove a rigidity lemma, that each prime–reciprocal sum is already in lowest terms and hence $P \cap Q = \varnothing$, and a barrier: any solution has $|P \cup Q| \geq 60$ and $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$, for both members and without hypotheses. This is the first valid unconditional bound of its kind; the only prior candidate rests on an invalid step, for which we give an explicit model. The barrier is uniform in the period, so it bounds the whole Ufnarovski–Åhlander cycle conjecture and not only its period–two case, and a parity dichotomy shows that an all–odd cycle would need at least 1412 primes.

Each level of the problem is finite and effectively decidable: for every pair of cardinalities there are only finitely many solutions, so the infinitude of admissible supports above level 59 bounds the search space and not the solution set. At level 61, the first open one, $\prod(P \cup Q) < 10^{567}$. No fixed–shape family of any growth rate can produce a solution, which closes the Thābit–Euler programme that is the classical technique for problems of this kind.

For the coprime relaxation recorded with the problem we show that the solutions containing the element 1 are exactly the primary pseudoperfect numbers, so that case is equivalent to Erdős Problem #313; the 1–free case needs 1414 members and carries an obstruction #313 does not.

The results are formalised in Lean 4, with every declaration depending only on the standard axioms and no appeal to kernel–external evaluation. A companion note records the exploratory and negative material: the certificate programme, the function–field lens, the breeder form and the square sieve. The problem remains open.

1 Introduction

Let P, Q be finite, nonempty sets of (distinct) primes. Erdős Problem #307 [10], attributed to Barbeau's 1976 "Computer challenge corner" problem #477 [2] and listed in Erdős–Graham [3], asks:

do there exist finite sets of primes P, Q with $\left(\sum_{p\in P}\frac{1}{p}\right)\left(\sum_{q\in Q}\frac{1}{q}\right) = 1\,?$

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The problem is verifiable in the sense that a single example would settle it. No example is known. The published partial facts are a remark of J. Rosen on MathOverflow [9], that any solution needs at least 59 primes in $P \cup Q$, and the problem page [10], which asserts $|P \cup Q| \geq 60$.

The relation between the two is worth stating precisely, since this paper proves the second. The page's justification reads, in full, that P and Q are disjoint and $\sum_{p \in P \cup Q} 1/p \geq 2$, "whence $|P \cup Q| \geq 60$". That implication does not hold as displayed: the reciprocals of the first 59 primes already sum to $2.00235 > 2$, so a mass of 2 is attainable on 59 primes and the stated hypotheses yield $|P \cup Q| \geq 59$, which is Rosen's bound. Reaching 60 requires excluding the 49,961 admissible 59–element supports one at a time, which is what Proposition 4.5 does and what `erdos307_sixty` machine-checks. We therefore claim no priority for the *statement* $|P \cup Q| \geq 60$, which predates this work on the problem page, and we do claim the proof. The known "solutions" of Cambie and others solve only the *weakened* version in which 1 is permitted as an element, e.g. $(1+\frac{1}{5})(\frac{1}{2}+\frac{1}{3}) = 1$; these do not bear on the prime version.

Expanding the product exhibits #307 as a rigidly shaped Egyptian fraction for 1: $1 = \sum_{p \in P} \sum_{q \in Q} 1/(pq)$, whose denominators are the *rectangular grid* $P \times Q$ of semiprimes; a *rank–one*, multiplicatively separable representation. The unrestricted companion (representing 1 by reciprocals of distinct two–prime–factor numbers, with no product structure imposed) is by contrast *solved*, and comprehensively so: every natural number is a sum of distinct semiprime unit fractions [24], and every a/b with b squarefree above an explicit threshold is likewise representable, so the $\omega = 2$ problem is settled well past the single value 1. Johnson (1978) gave a 48–term decomposition, Watanabe [5] 47–term ones, and Graham's survey [4] records exactly this contrast, noting after Barbeau that the product form "is not known." Thus #307 is precisely the rank–one restriction of a solved problem. Nor is the density–zero nature of the prime denominators the obstruction in itself: Bloom [6] represents every positive rational by distinct unit fractions drawn from the (equally sparse) shifted primes $\{p-1\}$. What resists is the rank–one rigidity (the demand that the representation *factor* as a product of two prime–reciprocal sums) which the barrier (Theorem 4.3) shows costs at least 59 primes, hence of order 870 semiprime terms against Johnson's 47.

Our starting point is an identification of #307 with a question about the *arithmetic derivative* $n \mapsto n'$, the unique map with $0' = 1' = 0$, $p' = 1$ for primes p, and the Leibniz rule $(ab)' = a'b + ab'$. (The map goes back to J. Mingot Shelly (1911) and appeared as a 1950 Putnam problem; Barbeau [1] gave the first systematic study, with the modern account in Ufnarovski–Åhlander [7] and OEIS [11].) For $n = \prod_i p_i^{e_i}$ one has $n' = n \sum_i e_i/p_i$. We show (Section 3) that #307 is *equivalent* to the existence of a two–cycle of $n \mapsto n'$ that is not a fixed point. The two–cycle reformulation itself is not new, and it is older than is generally recorded. Ufnarovski–Åhlander [7] prove that a period–two point of $n \mapsto n'$ has squarefree members with disjoint supports, and deduce that the period–two case of their Conjecture 3 is equivalent to their Conjecture 4: $\left(\sum_{i \leq k} 1/p_i\right)\left(\sum_{j \leq l} 1/q_j\right) = 1$ has no solution in distinct primes. That is #307, in the derivative literature, in 2003. It was restated "in disguise" by Seva [19] in 2019 (via $Q \mapsto Q \sum_{p|Q} 1/p$, which agrees with the arithmetic derivative on squarefree integers, where the question lives), who is the first to name it as #307, and obtained independently by Bado [26]; see Section 10. What is new here is the identification of the two problems — #307 is Ufnarovski–Åhlander's Conjecture 4 — and the resulting two–way bridge, importing their period–one conjecture and exporting our bounds. Remarkably, Barbeau authored foundational papers on both sides (1961 and 1976 [1, 2]).

The bridge is productive in both directions.

- *Importing* from the derivative literature: the Ufnarovski–Åhlander theorem that any two–cycle has squarefree members with disjoint supports [7] makes the "finite sets of distinct primes" hypothesis of #307 automatic, and links #307 to the Ufnarovski–Åhlander conjecture that all derivative cycles have period one.

- *Exporting* to that literature: our barrier (Theorem 4.3) is, we believe, the first valid unconditional bound of its kind. Kovič [8] states three bounds on an $n'' = n$ solution. A computer search gives $x > 10^4$, unconditionally. Proposition 17 gives $> 3 \cdot 5 \cdots 29 \approx 3.23 \times 10^9$, but only when the *smaller* member is odd; Kovič notes explicitly that no estimate follows otherwise. Proposition 18 claims the unconditional $r + s \geq 34$ and $\max\{m, n\} \geq \prod_{i=1}^{17} p_i \approx 1.92 \times 10^{21}$; its proof does not establish them, for the reason set out in Section 10. Against the strongest *valid unconditional* bound, 10^4, our $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$ is a gain of 52 orders of magnitude; against the conditional Proposition 17 it is 47, in the case where that proposition applies. On cardinality there is no prior bound to improve: $|U| \geq 60$ appears to be the first, and it re-derives Kovič's claimed $r + s \geq 34$ as a corollary by an unrelated route.

Results and organisation. Section 2 proves the Rigidity Lemma (Theorem 2.1): the fraction $\sum_{p \in S} 1/p$ is automatically in lowest terms, which forces, for any solution, the exact identities $N_P = D_Q$, $N_Q = D_P$, disjointness $P \cap Q = \varnothing$, and $D_Q = s\, D_P$ where $s = \sum_{p \in P} 1/p$. The disjointness core was anticipated by R. Israel [9]; the full rigidity is what powers everything later. Section 3 establishes the bridge. Section 4 proves the barrier, the parity dichotomy, and that longer cycles are strictly harder (so two–cycles are extremal). Section 2 of the companion gives a local congruence anatomy, a heuristic model, and a proof (via Erdős–Wintner) that the relevant abundance density is a genuine constant. Section 3 of the companion adds a function–field lens: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction in $\mathbb{Z}$ is purely archimedean. Section 5 develops the Frame Rule and divisor–trick, and, crucially, records the exact identity that renders the natural constructive objective circular. Section 4 of the companion recasts the rule as a bilinear breeder, records a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis, and shows quantitatively why (H) does not amount to a feasible finite search. Section 6 documents the computations, and Section 7 the Lean 4 formalisation. We are explicit throughout about what is proved, what is heuristic, and what is borrowed; the problem remains open.

Notation. For a finite set S of distinct primes put

$$D_S := \prod_{p \in S} p, \qquad N_S := \sum_{p \in S} \frac{D_S}{p}, \qquad \text{so} \qquad \sum_{p \in S} \frac{1}{p} = \frac{N_S}{D_S}.$$

We call N_S the *cofactor sum*. For a solution we write $a = D_P$, $b = D_Q$, $s = N_P/D_P = \sum_{p \in P} 1/p$ and $t = N_Q/D_Q = \sum_{q \in Q} 1/q$, so $st = 1$, and (without loss of generality) $s > 1 > t$. We write $\omega(n)$ for the number of distinct primes dividing n.

2 The Rigidity Lemma

Lemma 2.1 (Rigidity / lowest terms)**.** *For a finite set S of distinct primes,* $\gcd(N_S, D_S) = 1$. *Equivalently, $\sum_{p \in S} 1/p$ is already in lowest terms, and $v_p\big(\sum_{r \in S} 1/r\big) = -1$ for every $p \in S$.*

Proof. The prime divisors of $D_S = \prod_{r \in S} r$ are exactly the elements of S, so it suffices to show $p \nmid N_S$ for each $p \in S$. Fix $p \in S$ and reduce $N_S = \sum_{r \in S} D_S/r$ modulo p. For $r \neq p$ the term $D_S/r = \prod_{u \in S \setminus \{r\}} u$ still contains the factor p, so $p \mid D_S/r$. The single surviving term is $D_S/p = \prod_{u \in S \setminus \{p\}} u$, whence

$$N_S \equiv \prod_{u \in S \setminus \{p\}} u \pmod{p}.$$

Each u in this product is a prime distinct from p, so $p \nmid u$; as p is prime it does not divide the product. Thus $N_S \not\equiv 0 \pmod p$, i.e. $p \nmid N_S$. As p ranged over all prime divisors of D_S, $\gcd(N_S, D_S) = 1$. The valuation statement is the same computation: $v_p(N_S/D_S) = v_p(N_S) - v_p(D_S) = 0 - 1 = -1$. □

Remark 2.2. Distinctness and primality are both essential: v_p of the cofactor of a repeated prime need not be -1, and the final step uses Euclid's lemma. The case $|S| = 1$ is consistent ($N_{\{p\}} = 1$).

Theorem 2.3 (Solution structure)**.** *Suppose P, Q are finite sets of distinct primes with $\frac{N_P}{D_P} \cdot \frac{N_Q}{D_Q} = 1$. Then*

$$N_P = D_Q, \qquad N_Q = D_P, \qquad P \cap Q = \varnothing,$$

N_P is squarefree with prime support exactly Q (and N_Q squarefree with support P), and $D_Q = s\,D_P$ where $s = N_P/D_P$.

Proof. Cross-multiplying, $N_P N_Q = D_P D_Q$. By Lemma 2.1 both N_P/D_P and N_Q/D_Q are in lowest terms. Since $N_Q/D_Q = D_P/N_P$ and the lowest–terms representative of a positive rational is unique, comparing N_Q/D_Q with D_P/N_P gives $N_Q = D_P$ and $D_Q = N_P$.

Disjointness (the Israel mechanism [9]). If $r \in P \cap Q$ then $r \mid D_P$ and, by $N_P = D_Q = \prod Q$, also $r \mid N_P$, but $\gcd(N_P, D_P) = 1$ forces $r \nmid N_P$, a contradiction. (Equivalently $v_r(s) = v_r(t) = -1$, so $v_r(st) = -2 \neq 0 = v_r(1)$.) Hence $P \cap Q = \varnothing$.

Finally $N_P = D_Q = \prod_{q \in Q} q$ is a product of distinct primes, hence squarefree with prime support Q; symmetrically for N_Q. And $s\,D_P = (N_P/D_P)D_P = N_P = D_Q$. □

The disjointness in Theorem 2.3 means a solution is genuinely a partition of $U := P \cup Q$, and $D_P D_Q = \prod_{p \in U} p$. The valuation form $v_p(\sum 1/p_i) = -1$ underlying disjointness is due to R. Israel [9], and appears also in Seva [19]; the cross–matching form ($N_P = D_Q$, $N_Q = D_P$) was obtained independently and concurrently by Bado [26, Thm. A] (Section 10).

3 The bridge to the arithmetic derivative

Lemma 3.1. *For squarefree m, the arithmetic derivative equals the cofactor sum: $m' = \sum_{p|m} m/p = N_{\{p:\,p|m\}}$. Moreover $\gcd(m, m') = 1$.*

Proof. For $m = \prod_{p|m} p$ squarefree, $m' = m \sum_{p|m} 1/p = \sum_{p|m} m/p$, which is exactly the cofactor sum of the prime set of m. Coprimality is Lemma 2.1. □

Theorem 3.2 (Bridge)**.** *The following are equivalent.*

(1) *Erdős #307 has a solution: finite prime sets P, Q with $\left(\sum_{p \in P} 1/p\right)\left(\sum_{q \in Q} 1/q\right) = 1$.*

(2) *The arithmetic derivative has a two–cycle: integers $a \neq b$ with $a' = b$ and $b' = a$.*

(3) *The equation $n'' = n$ has a solution n that is not a fixed point (i.e. $n' \neq n$, so n is not of the form p^p).*

(*The implication* (2) $\Rightarrow$ (1) *uses the Ufnarovski–Åhlander squarefreeness theorem for cycles; all other implications, and everything used in the barrier of Section* 4*, are unconditional; see Remark* 3.3.)

Proof. (1) ⇒ (2). Let $a = D_P, b = D_Q$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$, and by Theorem 2.3, $N_P = D_Q = b$ and $N_Q = D_P = a$. Thus $a' = b$, $b' = a$, and $a \neq b$ since $s \neq 1$ (a prime–reciprocal sum is never 1, being N/D in lowest terms with $D > 1$).

(2) ⇒ (1). Let $a' = b, b' = a$ with $a \neq b$. By the Ufnarovski–Åhlander analysis of two–cycles (cf. [7, 8]), both a and b are squarefree; their prime supports P, Q are then disjoint since $\gcd(a, b) = \gcd(a, a') = 1$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$; the cycle gives $N_P = b = D_Q$ and $N_Q = a = D_P$, so $\left(\sum_{p\in P} 1/p\right)\left(\sum_{q\in Q} 1/q\right) = \frac{N_P}{D_P}\frac{N_Q}{D_Q} = \frac{D_Q}{D_P}\frac{D_P}{D_Q} = 1$.

(2) ⇔ (3). A two–cycle $a \neq b$ gives $a'' = a$ with $a' = b \neq a$, so a is a non–fixed solution of $n'' = n$. Conversely a solution of $n'' = n$ with $n' \neq n$ yields the two–cycle (n, n'). The fixed points of $n \mapsto n'$ are exactly the numbers p^p [1, 7]. □

Remark 3.3 (Status of the cited input)**.** Part (2) ⇒ (1) uses that a derivative two–cycle has squarefree, support–disjoint members. Squarefreeness is the content of the Ufnarovski–Åhlander analysis of cycles (a non–squarefree n has $p \mid \gcd(n, n')$ for some p, and along $n \mapsto n'$ such common factors propagate, precluding return); disjointness then follows from $\gcd(a, a') = 1$. We use it as a cited theorem. Even *without* it, the implication (1) ⇒ (2) and the entire barrier below are unconditional, because Rigidity already delivers squarefree, disjoint P, Q from a #307 solution.

Formalised. `Erdos307.bridge_forward` is (1) ⇒ (2) and cites nothing; the distinctness $a \neq b$ is *derived* there, not assumed (`dprod_ne_dprod`). Support–disjointness is available separately as `disjoint_primeFactors_of_cycle`, also from Rigidity; it is not consumed by `bridge_forward`, which does not need it. `bridge_backward` is (2) ⇒ (1) with the Ufnarovski–Åhlander squarefreeness appearing as an explicit hypothesis, and `two_cycle_iff_double_fixed` is (2) ⇔ (3) for an arbitrary self–map, on *no* axioms at all. 0 `sorry` (`lean/Erdos307/Bridge.lean`).

Corollary 3.4. *If the Ufnarovski–Åhlander conjecture holds in the weak form "every cycle of $n \mapsto n'$ has period one," then Erdős #307 has answer* NO*. Conversely, a* YES *to #307 exhibits a period–two cycle and refutes that statement.*

4 The barrier

Lemma 4.1 (Strict AM–GM)**.** *If $s, t > 0$ with $st = 1$ and $s \neq 1$, then $s + t > 2$.*

Proof. $s + t - 2 = s + 1/s - 2 = (s - 1)^2/s > 0$ since $s \neq 1$. □

Lemma 4.2 (The number 59, after Rosen [9])**.** *Any finite set U of distinct primes with $\sum_{p\in U} 1/p > 2$ has $|U| \geq 59$. Moreover, among k–element prime sets, $\sum 1/p$ is maximised by the k smallest primes.*

Proof. Replacing any prime of U by a smaller prime not in U increases $\sum 1/p$, so the maximum over k–element sets is at the first k primes. The reciprocal sum of the first 58 primes is $1.99874\ldots < 2$ and of the first 59 primes is $2.00235\ldots > 2$, so a sum exceeding 2 needs at least 59 primes. □

Theorem 4.3 (Barrier)**.** *For any solution (P, Q) of #307, with $U = P \cup Q$:*

(a) $|U| \geq 59$ *and* $\displaystyle\prod_{p\in U} p \geq \Pi_{59} := \prod_{i=1}^{59} p_i \approx 8.77 \times 10^{112}$*;*

(b) $\min(\prod P, \prod Q) \geq \sqrt{\Pi_{59}/T_{59}} = 2.0929\ldots \times 10^{56}$*, where* $T_{59} = \sum_{i=1}^{59} 1/p_i = 2.00235\ldots$*. In particular both* $\prod P$ *and* $\prod Q$ *exceed* 2.09×10^{56}.

Consequently every $n'' = n$ solution has both members $> 2.09 \times 10^{56}$. The strongest valid unconditional bound previously available is the computer search $x > 10^4$ of [8], so this is a gain of 52 orders of magnitude; against the conditional Proposition 17 of that paper (3.23×10^9, when the smaller member is odd) it is 47. The unconditional $r + s \geq 34$ and $\max\{m, n\} \geq 1.92 \times 10^{21}$ claimed in Proposition 18 of [8] are not established by their proof (Section 10); the bound above does not rely on them and in fact implies the first.

Proof. (a) By $st = 1$, $s \neq 1$ and Lemma 4.1, $\sum_{p\in U} 1/p = s + t > 2$, so $|U| \geq 59$ by Lemma 4.2; since P, Q are disjoint (Theorem 2.3), $\prod_{p\in U} p = D_P D_Q$, and a product of ≥ 59 distinct primes is at least the product of the 59 smallest, Π_{59}.

(b) Take $D_P \leq D_Q$ to be the minimal side. By Theorem 2.3, $D_Q = s\, D_P$ with $s > 1$, hence

$$s\, D_P^2 \;=\; D_P\, D_Q \;=\; \prod_{p\in U} p \;=: \; R, \qquad \text{so} \qquad D_P^2 = \frac{R}{s}.$$

Since $s \leq s + t = T(U) := \sum_{p\in U} 1/p$ (as $t > 0$),

$$D_P^2 = \frac{R}{s} \;\geq\; \frac{R}{T(U)}.$$

It remains to bound $R/T(U)$ below over all admissible U. Among k–element prime sets the quotient $\left(\prod U\right)/\left(\sum_{p\in U} 1/p\right)$ is minimised by the k smallest primes (smallest numerator and, by Lemma 4.2, largest denominator simultaneously), and passing from 59 to more primes only increases it (the product grows multiplicatively while the reciprocal sum grows by a single small term). Hence the global minimum over admissible U is at $U = \{p_1, \ldots, p_{59}\}$, giving $R/T(U) \geq \Pi_{59}/T_{59}$. Therefore

$$D_P^2 \;\geq\; \frac{\Pi_{59}}{T_{59}} \;=\; \frac{\Pi_{59}^2}{N_{59}} \;=\; 4.3806\ldots \times 10^{112}, \qquad D_P \;\geq\; 2.0929\ldots \times 10^{56},$$

where $N_{59} = T_{59}\, \Pi_{59}$ is the integer cofactor sum of the first 59 primes. ☐

Remark 4.4 (Why the *minimum* is bounded, not only the maximum)**.** The product identity $D_P D_Q \geq \Pi_{59}$ alone gives only $\max(\prod P, \prod Q) \geq \sqrt{\Pi_{59}} \approx 2.96 \times 10^{56}$. The stronger statement (b), a lower bound on the *smaller* side, genuinely uses rigidity: it is the identity $D_Q = s\, D_P$ (equivalently $D_P^2 = R/s$) together with $s \leq T(U)$ that converts the product bound into a bound on D_P. We verified numerically that $\min_U R(U)/T(U)$ is attained exactly at the first 59 primes (the minimising configuration is unique), so the constant 2.09×10^{56} is sharp for this method.

Proposition 4.5 (Closing the 59–prime level)**.** *No solution of #307 has $|P \cup Q| = 59$. Hence $|P \cup Q| \geq 60$, $\prod_{p\in U} p \;\geq\; \Pi_{60} > 2.46 \times 10^{115}$, and*

$$\min\left(\prod P, \prod Q\right) \;\geq\; \sqrt{\Pi_{60}/T_{60}} \;=\; 3.5053\ldots \times 10^{57}.$$

Proof (finite verification). Let $U = P \cup Q$ with $|U| = 59$; as in Theorem 4.3(a), $T(U) > 2$. The candidate list is finite and complete: (i) every prime $p \leq 167$ lies in U; omitting p caps $T(U)$ at $T_{60} - 1/p < 2$, since the 59 largest prime reciprocals avoiding p are those of the first 60 primes less p, and $1/p > T_{60} - 2 = 0.00590\ldots$ exactly for $p \leq 167$; (ii) every element $p' \geq 281$ satisfies $T_{58} + 1/p' > 2$ (the other 58 reciprocals sum to at most T_{58}), so $p' \leq 787$. Thus U consists of the 39 primes up to 167 plus twenty primes from $(167, 787]$, and exactly 49,961 such sets have $T(U) > 2$. For each, a solution supported on U forces both $N' + 2N = (a + b)^2$ and $N' - 2N = (a - b)^2$ to be perfect squares, where $N = \prod U$ and $N' = a^2 + b^2$ (rigidity; the $k = 0$ case of Proposition 10.2);

one integer test that simultaneously excludes all 2^{58} splits of U. Exact–integer verification shows $N'+2N$ is a non–square for *every* one of the 49,961 candidates, by two independent implementations with different enumeration orders and prunings (`code/close59.py` and `hunt/close59.rs` in the repository). Hence no 59–element support closes, and $|U| \geq 60$. The bound on the minimal side repeats the proof of Theorem 4.3(b) with the minimum of R/T now over $|U| \geq 60$, attained at the first 60 primes: $D_P^2 \geq \Pi_{60}/T_{60} = \Pi_{60}^2/N_{60}$, where N_{60} is the cofactor sum of the first 60 primes. $\square$

Corollary 4.6 (The coprime variant inherits the closed level)**.** *Consider the weakened variant in which the elements of P and Q are merely pairwise coprime integers ≥ 2 (the open form of the variant, $1 \notin P \cup Q$). Any solution has $|P \cup Q| \geq 60$ elements, and $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$.*

Proof. Rigidity holds verbatim for pairwise–coprime families (the cofactor–sum argument uses only coprimality), so again $T(U) = s + t > 2$. The least prime factors $\ell(e)$ of the elements are distinct, and $\sum_e 1/e \leq \sum_e 1/\ell(e)$. Suppose $|U| = 59$ and some element e is not prime, so $e \geq \ell(e)^2$. (Formalised: `Erdos307.coprime_loss` is the per–element bound, `loss_antitone` its monotonicity, and `loss_at_277` the numeric consequence (`lean/Erdos307/Coprime60.lean`); the slack itself is the level–59 census.) If $\ell(e) \leq 277$, the loss is $1/\ell - 1/e \geq (\ell-1)/\ell^2 \geq 276/277^2 = 0.00359\ldots$, exceeding the maximal available slack $T_{59} - 2 = 0.00235\ldots$; if $\ell(e) \geq 281$, then $e \geq 281^2$ while the other 58 reciprocals sum to at most T_{58}, giving $T(U) \leq T_{58} + 281^{-2} = 1.99875\ldots < 2$. Either way $T(U) < 2$, a contradiction: at 59 elements every element is prime, and Proposition 4.5 applies verbatim. The product bound transfers because $\prod_e e \geq \prod_e \ell(e)$ and $T(U) \leq \sum_e 1/\ell(e)$, so the extremal ratio bound at the first 60 primes still applies. $\square$

The next bound is what Corollary 4.11 consumes: for a support written as a base together with one further prime, the two square conditions bound that prime in terms of the base alone. It is stated here in the form the level theory uses; its role in the level–by–level exclusion programme is taken up in the companion.

Proposition 4.7 (The Pythagorean tail prime is bounded)**.** *Let S be a finite set of primes, $D = \prod S$, $N = \sum_{p \in S} D/p$, $A = N + 2D$, $B = N - 2D$, and suppose $D < N$. Let q be a prime with $q \nmid 2D$ for which $Aq + D$ and $Bq + D$ are both perfect squares. Then there are integers m, c with $mc = 4D$ and*

$$q^2m^2 - 4Nq + c^2 - 4D = 0, \qquad \big(qm^2 - 2N\big)^2 = 4\big(AB + Dm^2\big),$$

so that $q = 2(N \pm k)/m^2$ with $k^2 = AB + Dm^2$ and $m \mid 4D$. Moreover m is even, *say $m = 2\alpha$ with $\alpha \mid D$ and $\beta = D/\alpha$, and then q is a root of*

$$\alpha^2 q^2 - Nq + (\beta^2 - D) = 0, \qquad \text{so} \qquad q = \frac{N + \sqrt{N^2 - 4D^2 + 4\alpha^2 D}}{2\alpha^2},$$

a quantity strictly decreasing in α. Hence the maximum is at $\alpha = 1$ and

$$q \;\leq\; \tfrac{1}{2}\Big(N + \sqrt{N^2 - 4D^2 + 4D}\Big) \;=\; tD + O(1), \qquad t = \tfrac{1}{2}\big(\sigma + \sqrt{\sigma^2 - 4}\big), \quad \sigma = \sigma(S) = N/D.$$

The largest admissible tail is therefore the base times the very ratio b/a that the mass forces. At level 59 this is $9.207 \times 10^{112} = 1.0497\, D$, a factor 7.63 below $4N$.

Proof. Write $x^2 = Aq + D$ and $y^2 = Bq + D$ with $x, y \geq 0$. Since $A - B = 4D$ *exactly*, $x^2 - y^2 = (A - B)q = 4Dq$, that is $(x + y)(x - y) = 4Dq$. The prime q divides exactly one of the two factors: were it to divide both it would divide $2x$ and $2y$, hence x and y as q is odd, hence $q^2 \mid 4Dq$ and

$q \mid 4D$, against $q \nmid 2D$. Write the factor divisible by q as qm and the other as c, so that $mc = 4D$ and, in either case, $2x = qm + c$. Multiplying by m,

$$2xm \;=\; qm^2 + mc \;=\; qm^2 + 4D,$$

and squaring against $x^2 = (N+2D)q + D$ gives $4m^2\big((N+2D)q + D\big) = q^2m^4 + 8Dqm^2 + 16D^2$. Substituting $16D^2 = m^2c^2$ and cancelling $m^2 > 0$ leaves $q^2m^2 - 4Nq + c^2 - 4D = 0$; completing the square and using $AB = N^2 - 4D^2$ gives the second display. For the parity, $x+y$ and $x-y$ differ by $2y$ and so have equal parity; were both odd their product would be odd, whereas it is $4Dq$. Hence both are even, and as q is odd, m is even. Finally $m^2 \geq 4$ and $c^2 \geq 0$ give $4Nq = q^2m^2 + c^2 - 4D \geq 4q^2 - 4D$, that is $Nq \geq q^2 - D$; if $q \geq N+1$ then $q^2 - Nq - D \geq q - D > 0$ because $D < N < q$, a contradiction, so $q \leq N$. (Using only $m \geq 1$ here gives $q \leq 4N$; the parity removes the factor four.) For the sharp form, $m = 2\alpha$ and $c = 2\beta$ with $\alpha\beta = D$ turn $q^2m^2 - 4Nq + c^2 - 4D = 0$ into $\alpha^2q^2 - Nq + \beta^2 - D = 0$, whose larger root is $\big(N + g(u)\big)/(2u)$ with $u = \alpha^2$ and $g(u)^2 = N^2 - 4D^2 + 4uD$. Its derivative in u has the sign of $2uD - g(N+g)$, and $4uD = g^2 - N^2 + 4D^2$ turns that into $-\big((g+N)^2 - 4D^2\big)/2$, negative since $N > 2D$. So the root decreases in α and is largest at $\alpha = 1$. □

Proposition 4.8 (The level–by–level programme terminates at 60)**.** *Let U be a finite set of primes with $\sum_{p\in U} 1/p > 2$ and let B be arbitrary. Then there is a prime $q > B$ with $q \notin U$ such that $U \cup \{q\}$ again has $\sum 1/p > 2$. Consequently, for every $n > 59$ there are infinitely many n–element prime supports of mass exceeding 2.*

Proof. The primes are infinite, so choose a prime q exceeding both B and $\max U$; then $q \notin U$ and $\sum_{p\in U\cup\{q\}} 1/p = \sum_{p\in U} 1/p + 1/q > 2$. Iterating from the first 59 primes, whose reciprocal sum is $N_{59}/P_{59} = 2.00235\ldots > 2$, produces admissible supports of every cardinality above 59, each containing a prime beyond any prescribed bound. □

Remark 4.9 (What the level barrier costs, and what it does not)**.** Proposition 4.5 closes level 59 by enumerating its 49,961 admissible supports, and Proposition 1.2 together with the search of Remark 1.11 closes level 60 by adjoining one prime to each. The obvious continuation is to enumerate the admissible 60–element supports and repeat. Proposition 4.8 says there is no such enumeration *of supports*.

The finiteness at level 59 is an accident of one inequality being tight: 59 primes reach mass 2 only just, at $2.00235\ldots$, so demanding mass > 2 from exactly 59 primes pins them near the smallest 59. Allow a sixtieth and the first 59 already carry the mass alone, leaving the sixtieth entirely free. The mass condition, having been met, constrains nothing further. The next proposition shows that the *solutions* are nevertheless finite at every level: the haystack is infinite, the needles are not.

Proposition 4.10 (Each level carries only finitely many solutions)**.** *Fix cardinalities $m, n \geq 0$ and rationals $c_1, c_2 \geq 0$. The equation*

$$\Big(c_1 + \sum_{a\in A} \frac{1}{a}\Big)\Big(c_2 + \sum_{b\in B} \frac{1}{b}\Big) = 1$$

has only finitely many solutions in sets A, B of integers ≥ 2 with $|A| = m$, $|B| = n$; in particular, for every (m,n) there are only finitely many pairs of prime sets with $\sigma(P)\sigma(Q) = 1$, $|P| = m$, $|Q| = n$, and the bound is effective. 0 `sorry` (`Erdos307.sols_finite`, `erdos307_level_finite`).

Proof. Induct on $m+n$; the constants absorb removed elements. Write u, v for the two sums and $\varepsilon = 1 - c_1c_2$. Expanding, any solution satisfies $c_1v + u(c_2+v) = \varepsilon$, and the left side is strictly positive for $m+n \geq 1$ (if $m \geq 1$ then $u > 0$ and $c_2 + v > 0$, else the product is 0; if $m = 0$ then $c_1(c_2+v) = 1$ forces $c_1 > 0$, $v > 0$), so $\varepsilon > 0$ is fixed by (c_1, c_2) alone. One of the two terms is at least $\varepsilon/2$.

If $u(c_2+v) \geq \varepsilon/2$: every $b \in B$ has $1/b \leq 1/2$, so $v \leq n/2$, whence $u \geq \varepsilon/(2c_2+n)$; since u is a sum of m terms, some $a_0 \in A$ has $1/a_0 \geq u/m$, that is

$$a_0 \;\leq\; \frac{m\,(2c_2+n)}{\varepsilon}.$$

Moving a_0 into the constant, $(A \setminus \{a_0\}, B)$ solves the equation with $(m-1, n, c_1 + 1/a_0, c_2)$, and the map is injective for fixed a_0. If instead $c_1v \geq \varepsilon/2$, then symmetrically some $b_0 \leq 2c_1n/\varepsilon$ moves into c_2. Either way the solution set injects into finitely many strictly smaller problems, and the induction closes. The bounds compound: each step divides by an ε whose denominator is at most the product of the elements already fixed, so the resulting height bound is doubly exponential in $m+n$; crude, but effective. ◻

Corollary 4.11 (The largest element of a support is explicitly bounded)**.** *Let U be the support of a two–cycle, $q = \max U$, $S = U \setminus \{q\}$, $D = \prod S$ and $N = \sum_{p \in S} D/p$. Then*

$$N < 2D \implies q \leq \frac{D}{2D-N}, \qquad\qquad N \geq 2D \implies q \leq \tfrac{1}{2}\Big(N + \sqrt{N^2 - 4D^2 + 4D}\Big).$$

In both regimes q is bounded by an explicit function of S alone, so the enumeration at each level is a finite tree whose branching is known in advance. 0 `sorry` *for the mass regime* (`Erdos307.max_le_of_mass_deficit`, `max_le_of_mass_deficit'`)*.*

Proof. The mass of a two–cycle support is ≥ 2 ($\sigma(P)+\sigma(Q) \geq 2\sqrt{\sigma(P)\sigma(Q)} = 2$), so $N/D + 1/q \geq 2$. If $N < 2D$ then $1/q \geq 2 - N/D > 0$ and the first bound follows. If $N \geq 2D$ then $D < N$ and $q \nmid 2D$, and both $Aq + D$ and $Bq + D$ are squares, being $D(U) + 2\prod U = (\prod P + \prod Q)^2$ and $D(U) - 2\prod U = (\prod P - \prod Q)^2$ under $\prod U = qD$ and $D(U) = D + qN$; the second bound is then Proposition 4.7 at $\alpha = 1$, the discriminant being nonnegative because $N \geq 2D$. ◻

Proposition 4.12 (The mass ladder: the 59 smallest elements are bounded at every level)**.** *Let U be a set of k primes with $\sum_{p\in U} 1/p \geq 2$, written $u_1 < \cdots < u_k$. Then for every j with $\sum_{i<j} 1/u_i < 2$,*

$$u_j \;\leq\; \frac{k-j+1}{2 - \sum_{i<j} 1/u_i} \;\leq\; \frac{k-j+1}{2 - T_{j-1}}.$$

The hypothesis holds for all $j \leq 59$, since $T_{58} = 1.99874\ldots < 2$. In particular, at every level $k \geq 59$,

$$u_{59} \;\leq\; \frac{k-58}{2 - T_{58}} \;=\; 793.67\ldots \times (k-58),$$

so $u_1, \ldots, u_{59}$ are bounded by a quantity linear in k; only $u_{60}, \ldots, u_k$ escape the mass, and those are covered by Corollary 4.11. 0 `sorry` (`Erdos307.mass_ladder_step`, `mass_ladder_step'`)*.*

Proof. The $k-j+1$ elements $u_j, \ldots, u_k$ carry reciprocal mass at least $r_j = 2 - \sum_{i<j} 1/u_i$, and each is at least u_j, so $r_j \leq (k-j+1)/u_j$, which is the first bound. For the second, $u_i \geq p_i$ gives $\sum_{i<j} 1/u_i \leq T_{j-1}$, and $r_j > 0$ exactly when $\sum_{i<j} 1/u_i < 2$, which $T_{j-1} \leq T_{58} < 2$ guarantees for $j \leq 59$. ◻

Remark 4.13 (What this gives at the first open level)**.** At level 61 the ladder reads $u_1 \le 30.5$, $u_{10} \le 103.7$, $u_{30} \le 201.1$, $u_{58} \le 808.0$, $u_{59} \le 2381.0$: the 59 smallest elements of any level–61 support lie among the first 353 primes, so positions $1, \dots, 59$ range over at most $\binom{353}{59} \approx 10^{67.9}$ configurations, with u_{60} and u_{61} then confined by Corollary 4.11. This replaces the doubly exponential height of Proposition 4.10 at this level by an explicit and much smaller count. It is of course still far beyond enumeration; the point is that the first open level now has a stated size rather than an abstract finiteness.

Two consistency checks. At level 59 the ladder gives $u_{59} \le 793.67\dots$, and `Erdos307.elt_le_795` bounds the elements of a 59–element support by 795 by an unrelated argument; the same value appears again as the mass regime of Corollary 4.11 at the base of the first 58 primes. Falsification: over 5,781 supports of mass ≥ 2 generated by perturbing initial segments, 0 violations of the ladder, and over 5,885 supports 0 violations of $u_{59} \le (k-58)/(2-T_{58})$ (`code/mass_ladder.gp`).

Corollary 4.14 (An explicit height at the first open level)**.** *Every solution of #307 at level* 61 *has* $\prod(P \cup Q) < 10^{567}$.

Proof. Write $U = S \cup \{u_{60}, u_{61}\}$ with S the 59 smallest elements, $D = \prod S$.

If $\sigma(S) \ge 2$ then S is one of the 49,961 admissible 59–element supports of Corollary 6.16, whose products satisfy $\log_{10} D \le 113.397$ (`code/maxprod59.rs`). Splitting $S = P_0 \sqcup Q_0$ and writing $M = \prod P_0$, $N = \prod Q_0$, $M' = D(P_0)$, $N' = D(Q_0)$, Theorem 5.1 determines the two remaining primes: $u = (MN' + N^2)/\Delta$ and $v = (M'N + M^2)/\Delta$ with $\Delta = MN - M'N'$. Here $\Delta = D\big(1 - \sigma(P_0)\sigma(Q_0)\big)$ is a positive integer, since $\sigma(P_0)\sigma(Q_0) < \sigma(P)\sigma(Q) = 1$, so $\Delta \ge 1$; and $MN = D$ exactly, so $MN' = D\,\sigma(Q_0) \le tD$ with $t = 1.10200\dots$ the window root at T_{61}. Hence $u, v \le D^2 + tD$, giving $\log_{10} u, \log_{10} v \le 226.79$ and $\log_{10} \prod U \le 113.397 + 2(226.79) = 566.98$.

If $\sigma(S) < 2$ then Proposition 4.12 bounds every element of S and a branch–and–bound over the admissible chains gives $\log_{10} D \le 113.526$ (`code/ladder_maxprod.rs`). The ladder extends one step, $u_{60} \le 2/(2 - \sigma(S)) \le 2D$ since $2 - \sigma(S) \ge 1/D$, and Corollary 4.11 then gives $u_{61} \le tD\, u_{60}$, so $\log_{10} \prod U \le 454.75$. The first case dominates. ☐

Remark 4.15 (What the height means)**.** Proposition 4.10 makes every level finite by an argument using only the two cardinalities, and its height compounds doubly exponentially: at level 61 it is a tower, not a number one can write. Corollary 4.14 replaces it by 10^{567}, because it uses the two square conditions and the cycle equations rather than the sizes alone. Together with the barrier $\prod(P \cup Q) \ge (2.09 \times 10^{56})^2 = 10^{112.6}$ this confines level 61 to the window

$$10^{112.6} \;\le\; \prod(P \cup Q) \;<\; 10^{567}.$$

Two remarks on what this is and is not. It is not an enumeration: 10^{567} is far past any search, and the reduction that matters is not the height but Theorem 5.1, which turns the two unknown primes into a *computed* pair once the split of the bottom 59 is fixed, so the free parameters at level 61 are the support and the split, not the primes. And it is not a route to the problem, since the level index remains unbounded; it is a statement about one level.

Remark 4.16 (What the window does not do)**.** It is worth recording what fails, since the natural guess is wrong. One might hope the mass window pins all but the top few elements of a support, leaving a bounded search. It does not, and the pinning weakens as the level rises: a prime p is forced into a k–element support exactly when $1/p > T_{k+1} - 2$, which forces 39 primes at level 59, 27 at level 60 and only 21 (all $p \le 73$) at level 61, leaving 40 of the 61 elements unconstrained by mass. The bound of Corollary 4.11 comes from the square conditions, not from the mass, and this is why: past level 59 the mass condition, once met, constrains nothing further. Consistency check: at the base of the first 58 primes the mass regime gives $q \le 793.67\ldots$, while `Erdos307.elt_le_795` bounds the elements of a 59–element support by 795 through an unrelated argument (`code/level_dichotomy.gp`; 400 bases, 0 failures). At level 61 the second regime applies at the extremal base and gives $q \le 1.1020\,D$ with $\log_{10} D = 117.84$.

Remark 4.17 (The level–by–level programme survives its own barrier)**.** Together with Proposition 4.8 this splits the situation cleanly: at every level above 59 the admissible supports are infinite, and the solutions are finite with an effective height bound. Restricted to any fixed level, #307 is therefore *decidable*: a terminating computation either exhibits the finitely many solutions or certifies that the level is empty. What is genuinely unbounded in the problem is only the level itself, so #307 is equivalent to the halting of one explicit search, which is the precise content of the problem page's VERIFIABLE tag. We record the honest scale: the bound of Proposition 4.10 at level 61 is astronomically beyond enumeration, so decidability in principle does not yield a computation in practice; the value of the proposition is structural. Validated on the inhomogeneous case $c_1 = 1$, where solutions are the known 1–containing coprime examples: the enumeration recovers $(\{5\},\{2,3\})$, $(\{41\},\{2,3,7\})$, $(\{1805\},\{2,3,7,43\})$ and their congeners, every one containing an element within the proven bound (`code/level_finite_check.gp`).

Two consequences, and the second is the one that matters. The searches of Remark 1.11, however far they are pushed in $\omega(m)$, bear on level 60 alone; widening them is not a route upward. And any further progress on #307 must be *non–enumerative*: exhausting supports one level at a time is not merely expensive beyond level 60, it is impossible, so the method behind every unconditional exclusion in this note is now spent. That is a no–go, not a resource limit.

Formalised in `lean/Erdos307/LevelBarrier.lean` (`admissible_extends`, `admissible_of_card_add`, `level_enumeration_terminates`) on the three standard axioms, with a non–vacuity witness exhibiting the first 59 primes as a set the hypotheses apply to.

Theorem 4.18 (Parity dichotomy)**.** *Let $a' = b, b' = a$ be a derivative two–cycle (so a, b squarefree,* $\gcd(a,b) = 1$*). Then exactly one of the following holds.*

(i) Mixed parity. *Exactly one of a, b is even; say $a = 2k$ with k odd. Then $b = a'$ is odd and $\omega(b)$ is even.*

(ii) All odd. *Both a, b are odd. Then $P \cup Q$ avoids* 2*, so reaching reciprocal sum* > 2 *needs* ≥ 1412 *odd primes, and* $\min(a,b) > 10^{2519}$*. Moreover $|P|$ and $|Q|$ are both odd.*

Proof. They cannot both be even, as $\gcd(a,b) = \gcd(a,a') = 1$. In case (i), write $a = 2k$, k odd squarefree; $b = a' = k + 2k'$ is odd. For odd squarefree $b = \prod_i q_i$ one has $b' = \sum_i b/q_i \equiv \omega(b) \pmod 2$ (each b/q_i is odd). Since $b' = a$ is even, $\omega(b) \equiv 0 \pmod 2$. In case (ii) the same congruence gives the cardinalities: $b = a' \equiv \omega(a) \pmod 2$ with b odd forces $\omega(a) = |P|$ odd, and symmetrically $|Q|$ is odd. (The same conclusion follows from Proposition 2.3 of [20] applied to the expansion $1 = \sum_{p \in P, q \in Q} 1/(pq)$, whose rank is $|P|\,|Q|$, with $n = 2$; it does not improve $|P \cup Q| \ge 1412$, since $|P| + |Q|$ is even either way.) The union of supports omits 2; the reciprocal sum of the first 1411 odd primes is < 2 while that of the first 1412 exceeds 2, so $|P \cup Q| \ge 1412$ and $\prod_{p \in P \cup Q} p \ge 10^{5040}$ (product of the 1412 smallest odd primes, by direct computation). Applying the same rigidity/ratio argument as in Theorem 4.3(b) (now with $2 \notin U$) gives $\min(a,b) \ge \sqrt{R/T(U)} > 10^{2519}$. The all–odd case is therefore vastly more constrained than the mixed case and, heuristically, empty. □

Proposition 4.19 (Longer cycles are harder; two–cycles are extremal)**.** *Every member of a k–cycle of the arithmetic derivative is squarefree, and for $k \le 3$ the members are pairwise coprime (each* *pair is consecutive). Their masses $s(a_i) = a_{i+1}/a_i$ multiply to 1, so by AM–GM $\sum_i s(a_i) \geq k$; for $k \leq 3$ this is the union's reciprocal sum, whence the support contains at least m_k primes, where m_k is least with $\sum_{i \leq m_k} 1/p_i \geq k$. Now $m_2 = 59$ and $m_3 = 361{,}139$ (last prime 5,195,977), so a three–cycle has at least 361,139 primes in its support and product exceeding $10^{2.26\times 10^6}$ (hence its largest member exceeds $10^{752{,}194}$); by Mertens m_k grows doubly exponentially. Two–cycles are thus the extremal, and only practically relevant; case for that measure. For $k \geq 4$ non–consecutive members may share primes, so the displayed bound is one on the* multiset *reciprocal sum; the union's own sum is bounded in Proposition 4.20.*

Proposition 4.20 (The barrier is uniform in the period)**.** *Let $a_1 \to a_2 \to \cdots \to a_k \to a_1$ be a cycle of the arithmetic derivative with $k \geq 2$ and $a_i \neq a_{i+1}$, and let $U = \bigcup_i \operatorname{supp}(a_i)$. Then*

$$\sum_{p \in U} \frac{1}{p} \;\geq\; \frac{k}{\lfloor k/2 \rfloor},$$

so $|U| \geq m(k)$, the least n with $T_n \geq k/\lfloor k/2 \rfloor$, and $\prod_{p \in U} p \geq \Pi_{m(k)}$. Since $k/\lfloor k/2 \rfloor \geq 2$ always, with equality exactly for even k,

$$|U| \;\geq\; 59 \quad \text{and} \quad \prod_{p \in U} p \;>\; 8.77 \times 10^{112}$$

for a cycle of every *period.*

Proof. Rigidity gives $\gcd(a_i, a_i') = 1$, and $a_{i+1} = a_i'$, so a_i and a_{i+1} are coprime. Hence for each prime p the set $S_p = \{i : p \mid a_i\}$ contains no two cyclically consecutive indices, i.e. it is an independent set in the cycle graph C_k, and therefore $|S_p| \leq \lfloor k/2 \rfloor$. Consequently

$$\sum_i \sigma(a_i) \;=\; \sum_{p \in U} \frac{|S_p|}{p} \;\leq\; \left\lfloor \tfrac{k}{2} \right\rfloor \sum_{p \in U} \frac{1}{p}.$$

The masses satisfy $\prod_i \sigma(a_i) = \prod_i a_{i+1}/a_i = 1$, so $\sum_i \sigma(a_i) \geq k$ by AM–GM. Combining the two displays gives the claim, and T is increasing. □

The bound is uniform rather than growing, and it is the union support that a search or a sieve sees. Odd periods are genuinely harder, even ones are not:

k	2	3	4	5	6	7	8	9
$k/\lfloor k/2 \rfloor$	2	3	2	5/2	2	7/3	2	9/4
$m(k)$	59	361,139	59	1,413	59	400	59	231

so $m(9) < m(7) < m(5)$: the sequence is not monotone in k, and $m(k) = 59$ *exactly* for every even k, not merely in the limit. The consequence worth naming is that Theorem 4.3 is not a statement about period two at all: every nontrivial cycle of the arithmetic derivative, of any length, carries at least 59 primes and a product past 8.77×10^{112}. The barrier therefore bounds the whole of Ufnarovski–Åhlander's Conjecture 3 and not merely its period–two case, which is #307.

Corollary 4.21 (What a bounded sweep rules out, for every period at once)**.** *Let $a_1 \to \cdots \to a_k \to a_1$ be a cycle with every member at most X. Then*

$$k \;\geq\; \frac{\log_{10} \Pi_{59}}{\log_{10} X} \;=\; \frac{112.94\ldots}{\log_{10} X}.$$

Proof. Proposition 4.20 gives $|U| \geq 59$ for a cycle of any period, so $\prod_{p \in U} p \geq \Pi_{59} > 8.77 \times 10^{112}$. Every prime of U divides some member, so $\prod_{p \in U} p$ divides $\prod_i a_i \leq X^k$. ☐

X	10^6	10^7	10^9	10^{12}	10^{18}	10^{30}
rules out every $k \leq$	18	16	12	9	6	3

The sweep of Proposition 6.9, now carried to 10^8, therefore already excludes every cycle of period at most 14 with all members in range, and not merely period two. This is the practical form of Proposition 4.20: because the barrier does not weaken with the period, a single bounded search settles a whole initial segment of Ufnarovski–Åhlander's Conjecture 3 at once, whereas the period–two reading of the same computation settles only $k = 2$.

Proposition 4.22 (The barrier as a function of period and smallest prime). *Let $a_1 \to \cdots \to a_k \to a_1$ be a cycle with $k \geq 2$, let U be the union of the supports, and suppose* $\min U \geq P$. *With $c(k) = k/\lfloor k/2 \rfloor$, let $M(k, P)$ be the least m for which the m smallest primes $\geq P$ have reciprocal sum at least $c(k)$. Then $|U| \geq M(k, P)$, and $\prod_{p \in U} p$ is at least the product of those m primes.*

Proof. Proposition 4.20 gives $\sum_{p \in U} 1/p \geq c(k)$, and among sets of m primes all $\geq P$ the reciprocal sum is maximised by the m smallest such primes. ☐

$M(k, P)$	$k = 2$	$k = 3$	$k = 5$	$k = 7$	$k = 9$
$P = 2$	59	361,139	1,413	400	231
$P = 3$	1,412	—	361,138	40,260	15,473
$P = 5$	40,259	—	—	—	1,266,094
$P = 7$	588,500	—	—	—	—

(dashes exceed $\pi(4 \times 10^7)$; `code/kcycle_uniform.gp`). The family contains three results proved elsewhere in this note as special cases, and reproduces each of them exactly: $M(2, 2) = 59$ is Theorem 4.3; $M(2, 3) = 1412$, with product $10^{5040.1}$ and hence $\min(\prod P, \prod Q) > 10^{2519}$, is the all–odd branch of Theorem 4.18; and $M(3, 2) = 361{,}139$ is Proposition 4.19. That three independent derivations land on the same integers is the check the family deserves.

Two readings. The parity dichotomy is not special to period two: an all–odd cycle of *any* even period needs 1,412 primes and a product past 10^{5040}, and of period 9 still needs 15,473. And the family supplies, in valid form, the conditional statements that Proposition 18 of [8] asserts but does not establish: for $\min U \geq 3$ that proposition claims $r + s \geq 57$ where the true bound is $M(2, 3) = 1{,}412$, and for $\min U \geq 5$ it claims $r + s \geq 110$ where the true bound is $M(2, 5) = 40{,}259$ — larger by factors of 24.8 and 366.

5 A Thābit–type rule and its honest limits

The bridge recasts the search for a solution as the search for a derivative two–cycle, and the derivative's Leibniz structure makes the cycle equations *linear* in the "closing" primes. This yields a constructive calculus analogous to Thābit ibn Qurra's rule for amicable pairs and to Euler's divisor method.

Theorem 5.1 (Frame Rule). *Let M, N be coprime squarefree integers with derivatives M', N', and set $\Delta := MN - M'N'$. Suppose*

$$p = \frac{MN' + N^2}{\Delta}, \qquad q = \frac{M'N + M^2}{\Delta}$$

are integers, both prime, each coprime to MN, with $p \neq q$. Then $a := Mp$, $b := Nq$ form a derivative two–cycle $a' = b$, $b' = a$; equivalently (P, Q) with P the prime set of Mp and Q that of Nq solves Erdős #307.

Formalised. ***`Erdos307.frame_cycle`*** *verifies both cycle equations from the two divisibility hypotheses, and* ***`frame_solve`*** *runs the derivation backwards, so the two quotients are* forced *rather than guessed and the rule is an equivalence. The Leibniz step $a' = M'p + M$ is* ***`csum_insert_prime`***. *What no algebra reaches is the primality of the two quotients* (`lean/Erdos307/Frame.lean`).

Proof. With p prime and $p \nmid M$, $a = Mp$ is squarefree and $a' = M'p + M$; similarly $b' = N'q + N$. The cycle conditions $a' = b$, $b' = a$ read

$$M'p + M = Nq, \qquad N'q + N = Mp,$$

a linear system in (p, q) with determinant $-(MN - M'N') = -\Delta$. Solving gives exactly $p = (MN' + N^2)/\Delta$ and $q = (M'N + M^2)/\Delta$; substituting back verifies both equations identically. (Both identities were checked symbolically.) $\square$

A two–primes–per–side variant ("Rule 2") is obtained identically: with $a = Mp_1p_2$, $b = Nq_1q_2$ and elementary symmetric functions e_1, e_2, f_1, f_2, the cycle equations are linear in the f_i,

$$f_2 = \frac{Me_1 + M'e_2}{N}, \qquad f_1 = \frac{MNe_2 - MN'e_1 - M'N'e_2}{N^2},$$

and a solution is realised whenever $f_1^2 - 4f_2$ is a perfect square with prime roots.

An Euler-style divisor trick. Fix N and let $M = M_0 r$ with r a new prime, so $M' = M_0' r + M_0$. Put $\Delta_0 := M_0 N - M_0' N'$.

Proposition 5.2 (Divisor trick). *With the above notation and $p_{\text{num}} := MN' + N^2$:*

(a) $\Delta(r) := MN - M'N' = r\,\Delta_0 - M_0N'$ (linear in r)*;*

(b) $\Delta_0\, p_{\text{num}} - M_0N'\,\Delta(r) = K$ *where* $K := \Delta_0 N^2 + (M_0N')^2$ *is* constant in r*;*

(c) *consequently* $\Delta(r) \mid p_{\text{num}} \Rightarrow \Delta(r) \mid K$*, and the converse holds provided* $\gcd(\Delta(r), \Delta_0) = 1$ (*equivalently* $\gcd(M_0N', \Delta_0) = 1$*, since* $\Delta(r) \equiv -M_0N' \bmod \Delta_0$)*;*

(d) *each positive divisor $d \mid K$ with $d \equiv -M_0N'$ $(\mathrm{mod}\ \Delta_0)$ yields a candidate $r = (d + M_0N')/\Delta_0$ and recovers $\Delta(r) = d$; this enumeration is complete (it captures every genuine solution). The resulting $p = p_{\text{num}}/d$ is integral when* $\gcd(M_0N', \Delta_0) = 1$*, and otherwise must be checked.*

Proof. (a) and (b) are direct expansions (verified symbolically; $\partial K/\partial r = 0$). (c): from (b), $\Delta(r) \mid p_{\text{num}} \Rightarrow \Delta(r) \mid \Delta_0 p_{\text{num}} = M_0N'\Delta(r) + K \Rightarrow \Delta(r) \mid K$. For the converse, $\Delta(r) \mid K$ gives $\Delta(r) \mid \Delta_0 p_{\text{num}}$, whence $\Delta(r) \mid p_{\text{num}}$ iff $\gcd(\Delta(r), \Delta_0) = 1$. (d) inverts (a). Excluded degeneracies: $\Delta_0 = 0$ (r undefined) and $K \leq 0$. $\square$

Thus, exactly as in Euler's amicable–pair machinery, one factors the single number K once and reads off candidates from its divisors lying in a fixed residue class modulo Δ_0; only the primality of r, p, q then remains. Two caveats deserve emphasis: the converse in (c) and the automatic integrality of p require $\gcd(M_0N', \Delta_0) = 1$ (it fails, e.g., for $M_0 = 2$, $N = 19 \cdot 47$, $r = 5$, where $\Delta(r) \mid K$ but $\Delta(r) \nmid p_{\text{num}}$), and the count of admissible classes is governed by Δ_0, not merely by $\tau(K)$.

When the closing side carries a *single* new prime, the rule collapses to a normal form that separates the two layers of the problem.

Proposition 5.3 (One–prime normal form; the exactness stratum). (a) *Erdős #307 has a solution if and only if there exist squarefree M, N with*

$$MN - M'N' = M^2,$$

such that $p := N'/M$ is a prime not dividing M. Indeed the equation forces $M \mid M'N'$, hence $M \mid N'$ by $\gcd(M, M') = 1$*, and $N = M + M'p$; then $a = Mp$, $b = N$ satisfy $a' = M'p + M = N$ and $b' = N' = Mp = a$ (coprimality and $a \neq b$ are automatic). Conversely every solution (a, b) and every prime $p \mid a$ yield such a pair with $M = a/p$, so each solution carries $\omega(a)$ witnesses. The problem thus splits into an* exactness *layer (the Diophantine equation) and a single* primality *layer (one prime value of the determined quantity N'/M).* (b) The exactness layer alone carries the full barrier. *Call (M, N) a* relaxed witness *if $MN - M'N' = M^2$ with $p = N'/M$ composite. The identity $\big(\sigma(M) + \frac{1}{p}\big)\sigma(N) = \frac{N}{Mp} \cdot \frac{Mp}{N} = 1$ holds for* every *witness, so by strict AM–GM ($\sigma(N) = 1$ being impossible) $\sigma(M) + \sigma(N) > 2 - \frac{1}{p}$. Rigidity forces three structural facts. The supports of M and N are* automatically *disjoint: a common prime q would divide $N' = Mp$ through $q \mid M$, against* $\gcd(N, N') = 1$*. Likewise* $\gcd(p, MN) = 1$*: a prime shared by p and M propagates to $N = M + M'p$ and lands in the previous case, while one shared by p and N divides N'. And if p is even, both members are odd: $2 \mid N' = Mp$ forces N odd by the same lemma, while M even would make $N = M + M'p$ even. The union of supports therefore avoids every prime divisor of p, and for even p avoids 2; exact greedy computation then gives, for* every *composite p,*

$$\omega(MN) \;\geq\; 59, \qquad MN \;>\; 8.77 \times 10^{112}$$

the cycle barrier itself; with strictly stronger floors at every finite p: $\omega \geq 194$, $MN > 10^{497}$ at $p = 9$; $\omega \geq 230$, $MN > 10^{609}$ at $p = 4$; the even–p floors rising towards the all–odd bound 10^{5040}, and the infimum 8.77×10^{112} approached only as $p \to \infty$ through composites free of prime factors ≤ 277. Deleting primality thus lowers the barrier by nothing*: the 59–prime, 10^{112} wall is carried entirely by the exactness equation, and the primality of N'/M is a separate demand on top of it. (No witness, prime or composite, exists with $M \leq 120$, $p \leq 2000$, exhaustively.)*

The decisive obstruction: a circular objective. The heuristic yield of the divisor trick from one frame is

$$\mathbb{E}[\text{cycles}] \;\approx\; A \cdot \frac{1}{\ln r \, \ln p \, \ln q}, \qquad A = \#\{d \mid K : \; d \equiv -M_0 N' \pmod{\Delta_0}\} \;\approx\; \frac{\tau(K)}{\Delta_0},$$

the last step being an equidistribution heuristic. To make $\mathbb{E} \geq 1$ at the barrier scale ($1/\ln^3 \approx 5 \times 10^{-7}$) one needs $A \gtrsim 2 \times 10^6$ admissible divisors per frame, hence $\Delta_0 = O(1)$ with $\tau(K) \sim 10^6$. The natural objective is therefore "engineer frames with Δ_0 small." But here is the obstruction, an exact identity:

$$\boxed{\Delta_0 \;=\; M_0 N\big(1 - s_{M_0}\, s_N\big), \qquad s_{M_0} = \sum_{p \mid M_0} \frac{1}{p}, \quad s_N = \sum_{p \mid N} \frac{1}{p}.}$$

So Δ_0 is small *relative to* $M_0 N$ exactly when $s_{M_0} s_N \approx 1$ (which is the Erdős #307 condition for the frames themselves) and $\Delta_0 = O(1)$ in absolute terms at scale requires $s_{M_0} s_N = 1 - O(1/M_0 N)$, i.e. a reciprocal–product within $O(1/M_0 N)$ of 1, *strictly harder* than #307. The frame–engineering objective is thus circular: the Frame Rule is a faithful *parametrisation* of solutions, not a reduction in difficulty.

Remark 5.4 (Honest status of the constructive program)**.** The Frame Rule (Theorem 5.1) and divisor trick (Proposition 5.2) are exact theorems; they make the existence of a solution *falsifiable in principle by finite computation per frame*, and they place #307 in the same constructive lineage as Thābit's and Euler's rules. They do not, as they stand, lower the difficulty below #307 itself: by the boxed identity the bottleneck (small Δ_0) is the original problem in disguise, and a complete small–frame sweep (2.2×10^5 frames) produces zero cycles with total expected count < 1, as the barrier demands. We regard the calculus as an exact parametrisation of solutions whose central obstruction we have now identified precisely; not a pathway that lowers the difficulty, and not evidence that a solution is within reach.

Proposition 5.5 (No polynomial families)**.** *Let $p_1(t), \dots, p_k(t)$, $q_1(t), \dots, q_l(t)$ be integer polynomials, each taking* positive *prime values on a common infinite branch $t \to +\infty$ (a fixed–shape Schinzel–type family), such that $\left(\sum_i 1/p_i(t)\right)\left(\sum_j 1/q_j(t)\right) = 1$ for infinitely many t. Then every p_i and q_j is constant: the family is a single fixed solution. Consequently no polynomial parametrisation can reduce #307 to Bunyakovsky, Schinzel's Hypothesis H, or Bateman–Horn.*

Formalised. *`Erdos307.const_prod_eq_one_of_tendsto` is the limit step, `eps_eq_zero_of_identity` the vanishing step, and `nonconstant_empty` the conclusion that both index sets of nonconstant members are empty; `nopoly` chains them. The polynomial input enters only as the two hypotheses it supplies, that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to* 0*, so what is imported is visible.* 0 *`sorry`, three standard axioms* (`lean/Erdos307/NoPoly.lean`)*.*

Proof. Holding at infinitely many t, the relation holds identically as rational functions. A nonconstant p_i taking infinitely many prime (hence positive) values has positive leading coefficient on an infinite branch, so $p_i(t) \to +\infty$ there; split each side into constants and nonconstant members, $\sum_i 1/p_i = A_P + \varepsilon_P(t)$ with $\varepsilon_P(t) \to 0^+$. Letting $t \to \infty$ in the identity gives $A_P A_Q = 1$ (in particular both sides contain constants), and subtracting, $A_P\varepsilon_Q(t) + A_Q\varepsilon_P(t) + \varepsilon_P(t)\varepsilon_Q(t) = 0$ for all large t. Every term on the left is nonnegative, and strictly positive as soon as one nonconstant member exists; hence there are none. □

6 Computations

All verdicts use exact integer/rational arithmetic; floating point is used only for pre–screening.

- *Exhaustive non–existence below* 10^8*.* Enumerating all prime sets P with $\sum 1/p > 1$ and $\prod P \leq 10^8$ (forcing Q as the prime support of N_P and testing $N_Q = D_P$): 1,075,419 candidate sets, no solution. Subsumed by Theorem 4.3 but an independent sanity check.

- *Derivative sieve.* No solution of $n'' = n$ with $n' \neq n$ and both members $\leq 2 \times 10^7$ (smallest–prime–factor sieve of the derivative).

- *Function field.* The arithmetic derivative on $\mathbb{F}_q[t]$ has no two–cycles and no nonzero fixed points, verified exhaustively over $\mathbb{F}_2$ (degree ≤ 7), $\mathbb{F}_3$ (≤ 5), $\mathbb{F}_5$ (≤ 4), confirming Theorem 3.1.

- *Density.* The sieve gives density 0.042029 to 2×10^7 (0.017623 among squarefree); the independent Erdős–Wintner convolution over primes $\leq 10^5$ gives $\delta = 0.04202$, agreeing to four places (Proposition 2.6).

- *Conditional mass.* Of 43,087 qualifying squarefree a sampled, $b = a'$ is squarefree 95.0% of the time; on the resulting 40,920 squarefree b the mean reciprocal mass is 0.047 vs. the required $1/s \approx 0.934$, with empirical tail probability $\Pr[\sum_{q|b} 1/q \geq 1/s] = 0/40{,}920$.

- *Frame/ledger sweep.* Over 2.2×10^5 factorable small frames: the divisor–trick identities hold on every instance; the smallest Δ_0 observed is 22; the equidistribution $A \approx \tau(K)/\Delta_0$ holds only in aggregate ($\sum A = 127$ vs. $\sum \tau(K)/\Delta_0 = 39.8$) and fails per frame ($A = 0$ for 99.95% of frames); total expected cycles < 1; zero cycles, as the barrier requires.

- *μ–Sondow line census.* Enumerating the lines $n' = n \pm d$ (squarefree members coprime to d) finds no adjacent opposite–sign pair at additive distance d: exhaustively to 2×10^7 for $|d| \leq 200$, and along the small–$|\delta|$ genealogy tree out to members of more than a thousand digits. Every $\delta = \pm 1$ member encountered is a known Giuga or primary pseudoperfect number; the closest opposite–sign approach recorded is $|y - (x + d)| = 11$ (at $d = 1$, 30 vs. 42, and $d = 37$, 210 vs. 258), attained only among small members; the two trees do not come close at height.

Code and logs accompany this note.

7 Formalisation in Lean 4

The rigidity and barrier results are formalised against mathlib (Lean 4, toolchain `v4.30.0`). The following are machine–checked and `sorry`–free, depending only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`:

- Lemma 2.1 (`rigidity_coprime`) and Theorem 2.3 (`solution_structure`);

- Lemma 4.1 (`reciprocal_sum_gt_two`); the exact thresholds $\sum_{\text{first }58} 1/p < 2 < \sum_{\text{first }59} 1/p$ grounding $|U| \geq 59$ (`recipSum58_lt_two`, `recipSum59_gt_two`); the numeric core $\Pi_{59}^2 \geq (4 \times 10^{112})N_{59}$ (`barrier_numeric`), and the algebraic core of the barrier (`barrier`), giving $D_P^2 \geq 4 \times 10^{112}$;

- Theorem 6.30 (`log_lt_half_sub`, `delta_decreasing`, `two_le_add_of_mul_eq_one`) and Theorem 6.34 (`criterion_fails_of_pos`, `criterion_fails_of_nonpos`, `no_lambda_works`, `lyapunov_criterion_fa`
neither of which uses `native_decide`;

- Proposition 6.37 (`half_works`, `no_f_above_two`, `mass_sum_threshold`) and Proposition 6.38 (`exists_f_of_injective`, `increment_iff`, `no_two_cycle_of_decreasing`), which together machine-check the closure of the sign branch in Theorem 6.32;

- Proposition 6.24 (`exists_block_sum_near`, `infimum_zero`);

- Theorem 6.32(1) in its mechanism (`Erdos307.Congruential`) and Proposition 7.28 (`live_slot_threshold`, `threshold_bracket`).

The single non–elementary ingredient, that the k smallest primes minimise $\prod$ and maximise $\sum 1/p$ among k–element prime sets, is isolated as an explicit hypothesis of `barrier` (the ratio bound $R/T(U) \geq \Pi_{59}/T_{59}$), so the dependency is transparent rather than hidden. It is no longer an open dependency: `Erdos307.Extremal` proves it (`prod_first_primes_le`, `recipSum_le_first_primes`, `hRatio_of_extremal`), which is why Theorem 4.3 appears above in the hypothesis-free form `erdos307_barrier_clo`

On the trust boundary: there is none beyond the logic. Every declaration in the development, without exception, depends only on `propext`, `Classical.choice` and `Quot.sound`; there is no `native_decide` site and no `ofReduceBool`.

The one that resisted was the pruned search behind Proposition 4.5. `dfs` compares rationals, and the Lean kernel cannot reduce a `Rat` comparison at all, since `Nat.gcd` is well-founded recursion rather than a GMP primitive; `decide` does not merely run slowly on it, it fails to evaluate. Three changes move the execution into the kernel. Scaling by $M = \prod(\texttt{forced39} \cup \texttt{pool99})$, a 327–digit integer, clears every denominator *exactly*: each $1/p$ becomes M/p and thr becomes the integer $M \cdot \text{thr}$, so the scaled search branches identically and the soundness theorem `dfs_sound` is reused unchanged. The leaf test replaces `Nat.sqrt` on a 130–digit number, some hundreds of structural steps, by a residue certificate: eleven moduli $\{8, 19, 167, 127, 149, 67, 59, 101, 73, 191, 41\}$, with their quadratic residues stored as bitmasks, cover all 49,961 supports, and a non–residue is a proof of non–squareness (`lean/Erdos307/NonSquare.lean`). Finally the product and cofactor sum are carried down the recursion rather than rebuilt at each leaf. The kernel then runs the whole search in about a minute (`dfsA_run`, `dfs_of_dfsA`).

Proposition 4.5's product bounds are proved in this paper and not in Lean: `erdos307_sixty` establishes $|P \cup Q| \geq 60$ alone.

8 Why every known coprime example uses 1

Corollary 4.6 shows the weakened variant is no smaller than #307 itself: with $1 \notin P \cup Q$ a solution needs 60 pairwise coprime elements, and at 59 every element is forced prime. What that leaves unexplained is the shape of the examples that *are* known — $1 = (1 + \frac{1}{5})(\frac{1}{2} + \frac{1}{3})$, and the family generated by primary pseudoperfect numbers such as $1806 = 2 \cdot 3 \cdot 7 \cdot 43$ giving $1 = (1 + \frac{1}{1805})(\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{43})$. Every one of them uses the element 1. That is not an accident of search.

Proposition 8.1 (No coprime family has reciprocal sum 1)**.** *Let A be a finite set of pairwise coprime integers ≥ 2. Then $\sum_{a \in A} 1/a \neq 1$. 0* ***`sorry`*** *(`Erdos307.Coprime.sum_inv_ne_one`).*

Proof. Suppose $\sum_{a \in A} 1/a = 1$ and fix $a \in A$; put $R = \prod_{b \in A,\, b \neq a} b$. Multiplying by R gives $R/a + \sum_{b \neq a} R/b = R$. Each term of the sum is an integer, since $b \mid R$ for $b \neq a$, and so is R; hence $R/a \in \mathbb{Z}$, i.e. $a \mid R$. But a is coprime to every $b \neq a$, so $\gcd(a, R) = 1$ and $a = 1$, contradicting $a \geq 2$. ☐

Remark 8.2. Every known example of the weakened variant has one factor equal to $1 + 1/n$ and the other equal to $n/(n+1)$, i.e. one side is a coprime family whose reciprocal sum is $1 - 1/(n+1)$ and which is completed to 1 only by adjoining the element 1. Proposition 8.1 says the adjunction is unavoidable: the element 1 contributes a full unit to the sum while dividing nothing, and it is the only element that can, because for every other element the divisibility above closes. So the known examples are not near misses for the open form — they are on the other side of a divisibility, and no refinement of them approaches $1 \notin P \cup Q$.

The rigidity step that Corollary 4.6 inherits from the prime case is the same divisibility in cofactor form: for a coprime family A with product α, the numerator $M = \sum_{a \in A} \alpha/a$ satisfies $\gcd(M, a) = 1$ for every $a \in A$, because every term but α/a is divisible by a while $\alpha/a = \prod_{b \neq a} b$ is coprime to it. This is `Erdos307.Coprime.coprime_numerator`, and it closes the one step of Corollary 4.6's proof that was previously carried in prose.

8.1 A rejection sieve that factors nothing

The obvious search for a 1–free coprime example fixes one side and reads off the other: a coprime family A with product α determines its partner $\beta = \sum_{a \in A} \alpha/a$, and B must then be a coprime factorisation of β with $\sum_{b \in B} 1/b = \alpha/\beta$. The recorded obstacle is that testing a candidate appears to require factorising β, which at the sizes Corollary 4.6 forces is out of reach. It does not.

Proposition 8.3 (A necessary condition, computable in small modular arithmetic)**.** *Let A be a coprime family with $\alpha = \prod A$, $x = \sum_{a\in A} 1/a$ and $\beta = \sum_{a\in A} \alpha/a$. If some coprime family B satisfies $\big(\sum_A 1/a\big)\big(\sum_B 1/b\big) = 1$, then*

$$y_{\max}(\beta) \;:=\; \sum_{p^e \,\|\, \beta} \frac{1}{p^e} \;\geq\; \frac{1}{x}.$$

Moreover $y_{\max}(\beta)$ is computable without forming or factorising β: for every prime power p^e with $p \nmid \alpha$, Lemma 7.1 gives $\beta \equiv \alpha \sum_{a\in A} a^{-1} \pmod{p^e}$, so each $v_p(\beta)$ is decided by arithmetic modulo p^e alone.

Proof. Write $y = \sum_B 1/b = N/\gamma$ with $\gamma = \prod B$ and $N = \sum_{b\in B} \gamma/b$. As in Remark 8.2, $\gcd(\beta, \alpha) = \gcd(N, \gamma) = 1$, so $xy = 1$ reads $\beta N = \alpha\gamma$ with $\beta \mid \gamma$ and $N \mid \alpha$, forcing $\gamma = \beta$ and $N = \alpha$. Hence B is a coprime factorisation of β and $\sum_B 1/b = \alpha/\beta = 1/x$. Every coprime factorisation is obtained from the factorisation into prime powers by merging blocks, and merging strictly decreases the reciprocal sum, since $1/(mn) < 1/m + 1/n$ for $m, n \geq 2$ (0 `sorry`, `Erdos307.Coprime.merge_loses`). Hence the prime–power factorisation is the maximiser and $1/x = \sum_B 1/b \leq y_{\max}(\beta)$. The congruence is Lemma 7.1 with $r = p^e$, legitimate because $p \nmid \alpha$. □

The test is therefore free: primes above a trial bound Z contribute below $1/Z$ apiece, so trial moduli up to 10^5 already pin $y_{\max}$ to within 10^{-5}, using no integer larger than 10^{10}. Verified against direct multiprecision evaluation of $v_p(\beta)$ over 200 random families and all $p \leq 500$: 13,065 valuations, 0 mismatches (`code/beta_smoothness.rs`, checked by the script in the same file's header).

What the sieve says is stark. Over 5,272 splits of the first 60 primes lying in the mass window $|xy - 1| \leq 10^{-2}$, the mean of $y_{\max}(\beta)$ is 0.1795 and its maximum is 0.8371, against a requirement of $1/x \approx 0.95$: *not one candidate survives* (`code/beta_smoothness.rs`). The mechanism is a tension between the two constraints. Reaching $x \approx 1$ obliges A to hold small primes; but $\gcd(\alpha, \beta) = 1$ bars every prime of A from dividing β, and $y_{\max}$ is carried almost entirely by the small primes. The mass window and the smoothness requirement pull on the same scarce resource.

The same tension defeats the complementary route, which uses Lemma 7.1 in the forward direction: $b \mid \beta$ is equivalent to $\sum_{a\in A} a^{-1} \equiv 0 \pmod{b}$, a subset–sum condition on A that again needs no factorisation, so one may try to *impose* the divisibilities β must satisfy. Annealing on the forced divisor $D(A) = \prod\{b \in B : \ b \mid \beta\}$ over ground sets of 60 to 70 primes reaches $\log_{10} D \approx 10$ against $\log_{10} \beta \approx 50$–$75$ in a window $|xy - 1| \leq 10^{-2}$, and degrades to $\log_{10} D \approx 3$ as the window tightens to 10^{-7}, below which no split is found at all (`code/split_forcing.rs`). Since a random split already satisfies $\sum_{b\in B} 1/b \approx 1$ of the congruences by chance, the search inside the mass window is performing at the random baseline: the freedom the congruence route needs is exactly the freedom the mass window removes.

8.2 Candidates that are smooth by construction

Both failures above share an order of operations: fix a ground set U, split it, and hope β comes out smooth. Reversing the order removes the obstruction. Choose the small primes B_0 that are to divide β, draw A from primes above $\max B_0$, and *impose* $\sum_{a\in A} a^{-1} \equiv 0 \pmod{q}$ for each $q \in B_0$. Two points make this work where the earlier routes did not. First, the conditions are congruences on A alone, so they can be solved exactly rather than sampled — we use meet–in–the–middle over disjoint swap pairs drawn from adjacent primes, so the mass moves by less than 10^{-4}. Second, and decisively, B is no longer the complement of a fixed ground set but whatever factors β; the rigid window $x+y=T(U)$ of Section 8.1 therefore does not apply, and the only surviving requirement is $x>1$.

Two obstructions appear and both are removable. Every pool prime is odd, so $a^{-1}\equiv 1 \pmod 2$ and the residue mod 2 is pinned to $|A|$ mod 2: no swap can move it, and $|A|$ must be even. And forcing $q \mid \beta$ does not prevent $q^2 \mid \beta$, in which case the block contributes $1/q^2$ in place of $1/q$ — at $B_0=\{2,\dots,41\}$ an unconstrained solution came out with $2^5 \parallel \beta$ and $3^3 \parallel \beta$, which alone costs 0.76 and fails the sieve. Imposing $q \parallel \beta$, tested by the same identity taken modulo q^2, repairs it.

Proposition 8.4 (The necessary condition is constructible)**.** *For every B_0 among the first 8 to 12 primes there are coprime families A of primes above $\max B_0$ with $x=\sum_A 1/a>1$ such that each $q\in B_0$ divides β exactly once. Such A satisfy $y_{\max}(\beta)\geq\sum_{q\in B_0}1/q>1>1/x$, hence pass Proposition 8.3.*

Verified in exact multiprecision arithmetic at $B_0=\{2,\dots,37\}$: $|A|=3290$, largest member 31,063, $\log_{10}\alpha=\log_{10}\beta=13{,}163.1$, $\gcd(\alpha,\beta)=1$, every forced prime exactly dividing β, and

$$y_{\max}(\beta)=1.5927\ldots\ \geq\ 1/x=0.99597\ldots$$

(`code/smooth_by_construction.rs`). Against 0 of 5,272 for the sampled search, the construction passes at every depth from 8 to 12; depth 13 needs modulus 3.0×10^{14} and more swap pairs than the 2^{24} table cap admits.

Remark 8.5. The two routes are worth separating. The *factorisation* obstacle is genuinely gone — both tests run in arithmetic modulo 10^{10} and never form β — and Proposition 8.3 is a sound, essentially free first pass that any future search should run. What replaces it is a harder obstacle: inside the mass window the candidates are not merely hard to test, they are uniformly far from the target, by a margin of 0.77 in $y_{\max}$ at the observed best. A search for a 1–free coprime example must therefore generate candidates that are smooth *by construction* rather than filtered for smoothness after the fact. Section 8.2 gives such a construction. What it does not give is a test of sufficiency: deciding whether some coprime factorisation of β sums to exactly α/β needs the factorisation of β, whose unfactored part in the verified instance has 13,151 digits. Factoring returns, but only at the last step, and it no longer blocks the generation of candidates.

8.3 The two–member side, and what every known example really is

The searches above treat both sides symmetrically. They need not be. Suppose one side has exactly two members, $A=\{u,v\}$. Then $\alpha=uv$ and $\beta=u+v$, so u and v are the roots of $z^2-\beta z+\alpha=0$: *A is recovered from (α,β) by a quadratic, with no factorisation at all.* The roots are automatically coprime, since $\gcd(u,v)$ divides $\gcd(u+v,uv)=\gcd(\beta,\alpha)=1$.

Proposition 8.6 (The two–member reduction)**.** *Let B be a coprime family, $\beta=\prod B$, and let $u\geq 1$ be an integer with $u<\beta$. Then $\{u,\beta-u\}$ and B satisfy $\bigl(\sum_A 1/a\bigr)\bigl(\sum_B 1/b\bigr)=1$ if and only if*

$$D(B)\ =\ u(\beta-u),\qquad \textit{that is}\qquad \sum_{b\in B}\frac{1}{b}=\frac{u(\beta-u)}{\beta}.$$

0 ***sorry*** (`Erdos307.Coprime.two_element_identity`).

Proof. $\sum_{a\in A}1/a=1/u+1/(\beta-u)=\beta/\bigl(u(\beta-u)\bigr)$, and $\sum_B 1/b=D(B)/\beta$. The product is $D(B)/\bigl(u(\beta-u)\bigr)$, which is 1 exactly when $D(B)=u(\beta-u)$. ☐

The value of u decides everything, and it identifies the known examples.

$u = 1$. The condition is $D(B) = \beta - 1$, which is precisely the *primary pseudoperfect* condition, and $A = \{1, \beta - 1\}$. This is $(1 + \frac{1}{5})(\frac{1}{2} + \frac{1}{3})$ from $\beta = 6$ and $(1 + \frac{1}{1805})(\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{43})$ from $\beta = 1806$; the Sylvester products $2, 6, 42, 1806$ are pseudoperfect because Sylvester's recursion maintains $\beta - D(B) = 1$ identically. So the whole recorded family of solutions is the case $u = 1$, and it contains the member 1 for the structural reason that u *is* that member.

This is an equivalence, not merely a supply of examples, and it links two of Erdős's problems. That a primary pseudoperfect number yields a solution of the relaxation is an observation of Cambie on the problem page; Proposition 8.6 supplies the converse, since it derives $D(B) = u(\beta - u)$ as a *criterion.*

Corollary 8.7 (#307's relaxation at $u = 1$ is #313)**.** *The two–member solutions of the coprime relaxation of #307 that contain the element* 1 *are exactly the primary pseudoperfect numbers:* $\big(\{1, \beta - 1\}, B\big)$ *solves the relaxation if and only if* $\beta = \prod B$ *satisfies* $\sum_{q|\beta} 1/q + 1/\beta = 1$*. Consequently Erdős Problem #313, the infinitude of primary pseudoperfect numbers [25], is equivalent to the statement that the relaxation of #307 has infinitely many two–member solutions containing* 1*.*

Proof. Proposition 8.6 at $u = 1$ reads $D(B) = \beta - 1$, and dividing by β gives $\sum_{q|\beta} 1/q = 1 - 1/\beta$, which is the primary pseudoperfect condition. Both directions are the same computation. $\square$

The identity $D(B) = u(\beta - u)$ also names the right generalisation. Writing ∂ for the arithmetic derivative, [23] calls (R, c) a *port* and says a squarefree B *fills* it when $cB - R\,\partial(B) = 1$; the case $R = 1$, $c = 1$ is #313. Our condition is $u\beta - \partial(\beta) = u^2$, that is, the port $(1, u)$ filled at value u^2 rather than 1. The 1–free case of the relaxation is therefore a port–filling problem one step outside the range that #313 occupies, which is where any machinery developed for #313 would have to be pushed.

Proposition 8.8 (Only the value one sustains itself)**.** *For squarefree B and $c \geq 1$ put $\Delta_c(B) = cB - \partial(B)$. Then* $\gcd(\Delta_c(B), B) = 1$*, and appending a prime $q \nmid B$ gives $\Delta_c(qB) = q\,\Delta_c(B) - B$. Consequently a step returns to the same value, $\Delta_c(qB) = \Delta_c(B)$, only when $\Delta_c(B) = 1$, and then $q = B + 1$.* 0 ***sorry*** (`Erdos307.port_fixed_value`).

Proof. A prime $p \mid B$ dividing $\Delta_c(B)$ would divide $\partial(B)$, contradicting Theorem 2.3; so $\gcd(\Delta_c(B), B) = 1$. The recursion is $\partial(qB) = B + q\,\partial(B)$. Fixing the value gives $\Delta_c(B)(q - 1) = B$, so $\Delta_c(B) \mid B$, and coprimality forces $\Delta_c(B) = 1$. $\square$

Remark 8.9 (Why #313 has a family and the 1–free case has none)**.** Proposition 8.8 explains the asymmetry between Corollary 8.7's two sides. At $u = 1$ the target value is 1, which is a fixed point of the port step: from $\Delta = 1$ the choice $q = B + 1$ returns $\Delta = 1$, and iterating from $B = 2$ gives $2, 6, 42, 1806, \ldots$, Sylvester's sequence, whose partial products are exactly the known primary pseudoperfect numbers. The family costs nothing because the recursion sits still.

At $u \geq 2$ the target is u^2, and by the proposition *no* value other than 1 is self-sustaining, so there is no orbit to sit on: every step must be steered, and the deficit Δ wanders. This is the structural content of the failed descent recorded earlier — over 399 seeds the recursion $\Delta \mapsto q\Delta - B$ never reached a divisor of $B + 4$, the smallest Δ attained being 2.7×10^6 (`code/two_side_descent.gp`). The search was not unlucky. Verified: 23,916 pairs (B, c) with $\gcd(\Delta_c(B), B) = 1$ and 0 failures, 14,920 checks of the recursion with 0 failures, and among all self-sustaining steps found in a sweep, every one had $\Delta = 1$ (`code/port_recursion.gp`).

The consequence for transfer is specific. Machinery built for #313 targets the value 1 and may use its stability; anything of that kind does not carry to $u \geq 2$, where the stability is provably absent. What does carry is the recursion itself, which is value–blind.

Proposition 8.8 is sharp at period one only. A two–step orbit $\Delta \mapsto q_1\Delta - B \mapsto \Delta$ is possible above the value 1, and requires exactly $\Delta \mid q_1 + q_2$ (0 `sorry`, `Erdos307.port_period_two`); at $c = 1$ such orbits occur, for instance $B = 33$ with $\Delta : 19 \mapsto 5 \mapsto 19$ at $q_1 = 2$, $q_2 = 17$. They are single events rather than families, since returning a second time needs a fresh pair: the condition ties the primes to B. In the regime that the 1–free case actually requires — $c = 2$ with B and both primes odd — a sweep of 162,110 states found none (`code/port_orbit.gp`).

Proposition 8.10 (The barrier survives relaxing the equations to divisibilities)**.** *Let P, Q be disjoint finite sets of primes and suppose only that*

$$\prod Q \;\Big|\; D(P) \qquad \textit{and} \qquad \prod P \;\Big|\; D(Q),$$

in place of the two equalities a two–cycle requires. Then $\sigma(P)\sigma(Q) \geq 1$, hence $\sigma(P) + \sigma(Q) \geq 2$, and therefore $|P \cup Q| \geq 60$ and $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$ as in Theorem 4.3. 0 *`sorry`* (`Erdos307.mass_from_divisibility`).

Proof. Divisibility between positive integers gives $\prod Q \leq D(P) = \sigma(P)\prod P$ and $\prod P \leq D(Q) = \sigma(Q)\prod Q$. Multiplying and cancelling the positive quantity $\prod P \prod Q$ leaves $1 \leq \sigma(P)\sigma(Q)$, and the arithmetic–geometric mean inequality gives $\sigma(P) + \sigma(Q) \geq 2$, which is the only input Theorem 4.3 consumes. □

Remark 8.11 (What the relaxation localises)**.** The relaxation is worth recording because it is natural to expect the mass condition to be an artefact of the *exact* equations, so that weakening them would open room below level 60. It does not: the divisibilities alone already force the mass, because a divisibility between positive integers is an inequality, and the two inequalities point in opposite directions. Consequently the difficulty in #307 is not located in the passage from "$\prod Q$ divides $D(P)$" to "$\prod Q$ equals $D(P)$"; both carry the same barrier.

At $|P| = |Q| = 2$ the relaxed system has no solutions at all among the primes below 283 (2,925,810 pairs, 0 solutions, `code/relax_divisibility.gp`), and Proposition 8.10 says why: four primes cannot carry mass 2. A random search for relaxed solutions is therefore vacuous rather than evidential, and the proposition is checked instead on the arithmetic it actually asserts, over 200,000 random positive quadruples with 0 counterexamples.

Corollary 8.12 (The balanced splits are exactly the forbidden ones)**.** *In the situation of Corollary 4.14, a split $S = P_0 \sqcup Q_0$ can support a solution only if $\sigma(P_0)\sigma(Q_0) < 1$, hence only if $\sigma(P_0)$ lies strictly* outside *the base's own mass window $[1/t_0, t_0]$, where $t_0 + 1/t_0 = \sigma(S)$. In particular the mass–balanced splits, $\sigma(P_0) \approx \sigma(S)/2$, are precisely excluded.* 0 *`sorry`* (`Erdos307.split_outside_window`).

Proof. Theorem 5.1 gives $\alpha = (MN' + N^2)/\Delta$ with $\Delta = D(1 - \sigma(P_0)\sigma(Q_0))$, and $\alpha > 0$ forces $\Delta > 0$. With $\sigma(P_0) + \sigma(Q_0) = \sigma(S)$ fixed, the product is the downward parabola $z \mapsto z(\sigma(S) - z)$, which exceeds 1 exactly between the roots of $z^2 - \sigma(S)z + 1$ and is maximal at the midpoint. □

Remark 8.13 (Why level 61 is still not enumerable)**.** This is the complement of Proposition 6.14, which places the mass of a *solution* inside the window of the full support; here the split of the *base* must sit outside its own. It is a genuine constraint and it points a search away from where one would naturally begin, but it does not make level 61 enumerable, and it is worth recording the arithmetic rather than leaving the size vague. Sampling 2×10^6 splits of the first 59 primes, 89.4% satisfy $\sigma(P_0)\sigma(Q_0) < 1$, so the condition removes about one split in nine (`code/split_balance.gp`). Against 49,961 bases and 2^{59} splits each, some 2.6×10^{22} candidates remain, and the product–balance $M \approx N \approx \sqrt{D}$ that Theorem 5.1 prefers trims this by a further factor of order 10, not of order

10^{10}. Each surviving candidate still needs two divisibility tests on 113–digit integers. We therefore record level 61 as out of computational reach by roughly ten orders of magnitude, and note that no constraint in this paper closes that gap.

Remark 8.14 (What actually obstructs, in both problems)**.** It is worth being exact about where the difficulty sits, because the port dynamics is not where it sits. Along Sylvester's chain the value is $\Delta \equiv 1$ *identically*, so for #313 the value dynamics never obstructs at all; what obstructs is that the next element $B/\Delta + 1$ must be *prime*, and the chain $2, 6, 42, 1806$ stops because $1807 = 13 \cdot 139$. That single composite is the whole reason #313 is open rather than settled by Sylvester.

The comparison is then clean, and it is the honest conclusion of the transfer. #313 carries one obstruction: the primality of a constructed element. The 1–free case of #307's relaxation carries that same obstruction *and*, by Proposition 8.8, a second one that #313 does not have — no self–sustaining value to run a chain along in the first place. So the two problems are not of equal difficulty under this correspondence: #307's relaxation is strictly harder in an identifiable way, and a solution of #313 would not by itself supply one. This is a negative conclusion, but it is a proved one rather than a failed search, and it retires the transfer as a route.

$u = 2$. The first 1–free case. The condition is $D(B) = 2\beta - 4$, i.e. $\sum_{b\in B} 1/b = 2 - 4/\beta$, and then $A = \{2, \beta - 2\}$, which needs β odd so that $\gcd(2, \beta - 2) = 1$ (0 `sorry`, `Erdos307.Coprime.two_free_case`). Reducing $D(B) = 2\beta - 4$ modulo each $b \in B$ gives the Znám–type system $\prod_{b'\neq b} b' \equiv -4 \pmod{b}$, and conversely those congruences force $\beta \mid D(B) - 2\beta + 4$, so with $1 < \sum_B 1/b < 2$ the quotient is an integer in $(-1, 0]$ and must vanish: *the local conditions imply the global one.*

Proposition 8.15 (u is the mass, so $u = 1$ is the only cheap case)**.** *In the situation of Proposition 8.6, $\sum_{b\in B} 1/b = u(\beta - u)/\beta = u - u^2/\beta$. Hence the reciprocal sum of B is u up to $O(u^2/\beta)$: the parameter u is the mass B must carry. Consequently the cost is monotone in u, and $u = 1$ — the case that contains the member 1 — is the only value for which B needs mass ≈ 1 and can therefore be short.*

This settles what one might hope for from larger u. Since $\sum_A 1/a = 1/u + 1/(\beta - u)$ and $\sum_B 1/b$ is its reciprocal, asking for $1 \notin P \cup Q$ is asking for $u \geq 2$, which is asking for B to carry mass at least $2 - 4/\beta$; $u = 3$ demands $3 - 9/\beta$, and so on. There is no cheaper u hiding above 2, and no analogue of Sylvester's invariant to be had there: the invariant $\beta - D(B) = 1$ works at $u = 1$ precisely because mass 1 is reachable in four terms. Verified exhaustively: over every $\beta \leq 4 \times 10^7$ and *every* coprime factorisation of it — all set partitions of its prime powers — the discriminant $\beta^2 - 4D(B)$ is a square with admissible root in exactly 7 cases, and every one has $u = 1$. The largest u attained anywhere in that range is 1 (`code/two_member_scan.rs`).

Remark 8.16 (The price of dropping the member 1)**.** The reduction is exact and factoring–free, but it fixes the size. Since $\sum_B 1/b = 2 - 4/\beta$ and B consists of odd pairwise coprime integers, the reciprocal sum must exceed 2, and $\sum_{3\leq p\leq X} 1/p$ first exceeds 2 at the 1412nd odd prime $p = 11789$ (sum $2.0000206\ldots$). Hence $|A \cup B| \geq 1414$, against 60 for the prime problem: passing from $u = 1$ to $u = 2$ costs a factor of 23 in the number of members, because the member 1 was carrying a full unit of mass that must now be found among odd reciprocals. This is the exact price of the gap on the problem page, and it is why the Sylvester descent, which produces the $u = 1$ family for free, does not reach $u = 2$: driving $e = 2\prod B - D(B)$ to a divisor of $\beta + 4$ by the recursion $e \mapsto be - \prod B$ failed over 399 seeds, the smallest e reached being 2.7×10^6, while the products outrun any bound long before 1412 members (`code/two_side_descent.gp`).

Remark 8.17 (Where the neighbouring literature stands, and what it supplies)**.** Three recent items bear on the problem without deciding it, and it is worth saying exactly what each does and does not give.

[20] studies $\sum_i 1/x_i = 1$ with the prime factors of the x_i confined to a fixed finite set S, motivated by the classification of integral fusion categories, and supplies algorithms for that restricted search. This is the computational problem a level of #307 presents, and the paper records the same obstruction Proposition 4.10 meets from the other side: the general bound on the solutions is doubly exponential in the number of terms. Its "low–rank" refers to the number of terms, not to the rank–one product structure that distinguishes #307 from the unrestricted problem; the two senses should not be conflated.

[21] treats an iterated arithmetic function of Erdős and Graham, $g(n) = n + \varphi(n)$, and settles the two–step case by reducing it to the single equation $\varphi(n) + \varphi(n + \varphi(n)) = n$, enumerating the solutions, and leaving a residue that requires a prime $p \geq 10^{10}$ with $\varphi((3p-1)/4) = (p+1)/2$, apparently absent. The shape is the shape of #307: a two–cycle for an iterated arithmetic function, reduced by an exact identity to one equation, with a residue that is a rarity condition. Theorem 5.1 performs the same reduction here. The difference is the scale of the residue, which the barrier places above 10^{56} rather than 10^{10}, and that difference is the whole of why the same method finishes there and not here.

Its algorithms do not reach #307: they are built for one or two distinct primes with high multiplicity (the paper says "a small number of primes, say 1 or 2", and its tables are for $\{p\}$ and $\{2, p\}$), whereas a solution here needs at least 60 distinct primes each to the first power. The two regimes are disjoint, not merely separated by scale. What does transfer is one necessary condition, recorded in Theorem 4.18.

[22] bounds the distance from 1 to the best reciprocal subsum of a multiset of reciprocal mass K by $\exp(-c\sqrt{K \log K})$, improving $\ll K^{-2}$. It concerns how well 1 can be approached, which is the direction Proposition 6.10 occupies, and says nothing about exact representation, which is what #307 asks.

9 Status and questions

Erdős #307 remains **open**. It is Conjecture 4 of Ufnarovski–Åhlander [7], so it is equivalent to the existence of a non–trivial two–cycle of the arithmetic derivative and has been open in that literature since 2003 under another name. We establish rigidity and a barrier, $|P \cup Q| \geq 60$ with $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$, which voids all direct search; it appears to be the first valid unconditional bound of its kind, the only prior candidate resting on an invalid step (Section 10). The rigidity result and the barrier are Lean–verified, extremality proved rather than assumed.

The barrier is one instance of a single inequality. Writing $n(c) = \min\{n : T_n \geq c\}$,

$$\sum_{p \in U} 1/p \ \geq \ c \implies |U| \geq n(c), \qquad \prod_{p \in U} p \ \geq \ \Pi_{n(c)},$$

and the arithmetic of each result enters only in producing its c (Remark 6.13). Read in the size it gives the double–logarithmic frontier law (Proposition 6.11); in the near–miss ratio, the cost family (Corollary 6.12), whose endpoint at $r = 1$ is the barrier and which shows $a'' \geq 2a$ needs 37,963 primes; in the period, that the barrier is uniform in k, so it bounds the whole of Ufnarovski–Åhlander's Conjecture 3 and not only #307 (Proposition 4.20); in the orbit length, that the derivative cannot outgrow its own support (Proposition 6.21); in the split, that $\min(\omega(a), \omega(b)) \geq 3$ whenever $|U| \leq 68$ (Proposition 6.20); and in the level, the forced core and the ceiling that pin level 59 from both sides and yield its admissible count 49,961 in closed characterisation (Propositions 6.14, 6.15, Corollary 6.16). The algebraic cores of these are Lean–verified (`Dictionary.lean`).

On the constructive side we give a Thābit/Euler–type calculus and a bilinear breeder whose candidate q is always integral (Proposition 4.2), a conditional reduction (H) $\Rightarrow$YES, and a quantitative ledger showing (H) is not a feasible finite search. No fixed–shape family of any growth rate can parametrise #307 (Proposition 4.4), which closes the Thābit/Euler programme rather than only its polynomial shadow. We also give a parity dichotomy, an Erdős–Wintner density theorem, and a function–field theorem localising the obstruction to the archimedean place. The heuristics favour YES, and the model behind them survives a second–moment test (Proposition 6.27); we claim no proof in either direction.

Open problems: (i) close the residual 3,478 single–tail families at level 60, and close the pair sector of Proposition 1.3. What this buys should be read with Proposition 6.18: the admissible supports do not form a tree under adjunction, so completing every one–new–prime family at level 60 would still not close level 60; for which Proposition 1.5 rules out every congruence kill, so only global, per–family input (factorisation or direct search) can serve (the canonical tail family is closed, Proposition 1.2; the finite verification of Proposition 4.5 is Lean–checked, `erdos307_sixty`; the conditional 3/4 law is explained; it is the reciprocity switch of Remark 1.6, made exact by the class–vector functional there); (ii) decide the parity dichotomy's all–odd case (we expect it empty); (iii) sharpen the certified enclosure $0.04202 \leq \delta \leq 0.04223$ of Proposition 2.6, whose width is carried almost entirely by the tail window a; three digits are certified, and a longer envelope convolution with a smaller window certifies more; (iv) settle the density hypothesis (H) of Section 4 of the companion, or exhibit a non–circular frame family escaping the $\tau(K)/G$ ceiling; (v) find a $\mathbb{Z}$–side structure detecting the archimedean obstruction of Section 3 of the companion; one that must be non–local, since by Proposition 2.2 no single modulus obstructs; (vi) settle #307.

Remark 9.1 (The place of the problem: a local–global dictionary)**.** The pieces above assemble into a dictionary. (i) For a fixed support U, the subset form of Proposition 4.3, $A(\Sigma - A) = D^2$, satisfies the *Hasse principle as an equation*: an integer solution A exists iff one exists over $\mathbb{R}$ and over every $\mathbb{Z}_\ell$ (the discriminant $\Sigma^2 - 4D^2$ must be a nonnegative square, and squares obey local–global). Its real–place condition is $\Sigma \geq 2D$, i.e. $\sigma(U) \geq 2$: *the barrier is precisely the real–place solvability condition of the subset form.* (ii) The barrier never needed primality: pairwise coprime integers have distinct smallest prime factors, so $\sum 1/a_i \leq \sum 1/p_i$, and *any* pairwise–coprime support of mass ≥ 2 has at least 59 elements and product at least Π_{59}. Coprimality and the ordering of $\mathbb{R}$ carry the wall; primality enters only through rigidity. (iii) What remains unsolved is the *realization* of the root $A^* = \left(\Sigma + \sqrt{\Sigma^2 - 4D^2}\right)/2$ as a sub–multiset sum of the cofactors D/p. Its local conditions are exactly the local anatomy of Lemma 2.1: modulo ℓ^2 for $\ell \in U$ the subset sums occupy precisely the two cosets $\ell\mathbb{Z} \cup (D/\ell + \ell\mathbb{Z})$ (2ℓ residues, observed exactly (6 of 9, 10 of 25, 14 of 49, 22 of 121)) and these are the same constraints the equation itself forces: no completion separates the equation from its realization. In the language of integral points, then: #307 is everywhere locally soluble, with the barrier as its archimedean height condition, and its resolution is a *strong approximation* question for the thin, prime–structured set of subset sums a failure mode invisible to Brauer–Manin–type obstructions, consistent with every no–go above and with the existence of cycles over the neighbouring rings. We record this as a dictionary, not a method.

Attribution. We are careful to separate prior, concurrent, and present contributions (details in Section 10). *Prior/concurrent:* the two–cycle reformulation, and its equivalence with the arithmetic derivative, are due to Ufnarovski–Åhlander [7] (2003, Conjecture 4); the naming of it as #307 to Seva [19] (2019); an independent rederivation to Bado [26]; the bound $|P \cup Q| \geq 59$ to

J. Rosen and the disjointness valuation $v_p(\sum 1/p_i) = -1$ to R. Israel [9]; the source identification to "Woett" [9]. Bado [26] independently obtained the rigidity cross–matching, the local anatomy, the parity restriction, a one–sided exact test, approximate solvability, and the union criterion of Proposition 10.1. *Present note:* the identification of #307 with Ufnarovski–Åhlander's Conjecture 4 and the resulting two–way bridge to the $n'' = n$ literature; the product barrier 2.09×10^{56}; the function–field theorem and archimedean localisation; the constructive deficit analysis and seed obstruction; the Erdős–Wintner density theorem, and the k–cycle barriers. Both external sources [19, 26] have been examined directly and the overlaps above reflect their actual contents.

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Remark 9.2 (Where a proof of NO would have to come from)**.** Four results of this note combine into a single statement about the shape any negative proof must take, and it is worth assembling because each of the four is easy to read as an isolated obstruction.

Proposition 1.9 shows the cycle system is soluble modulo every m, so it has solutions in $\widehat{\mathbb{Z}}$ and there is no congruence obstruction to find, at any modulus or any finite combination of moduli. Theorem 3.1 shows the analogue over $\mathbb{F}_q[t]$ is false, having no cycles at all, so no argument that survives the passage to function fields can decide #307 negatively: whatever a proof uses, $\mathbb{F}_q[t]$ must lack it. Proposition 5.11 shows no place carries a descent. Together these localise the obstruction at the archimedean place, which is the reading already recorded.

The fourth is the constraint. The only archimedean tool in play is the mass, and Proposition 1.10 shows it is exactly tight: raising the barrier one level requires $\sigma(a) > t_n$, the only unconditional bound available is $\sigma(a) - 1 \geq 1/a$, and $\{2, 3, 7, 41\}$ realises $\sigma = 1.00058\ldots$ on four primes, so nothing forbids $a' - a = 1$. So a proof of NO needs an archimedean argument that is not the mass, and we know of none. This is not a claim that none exists; it is a statement of what the four results jointly exclude, recorded so that the search is not repeated inside the region they cover.

10 Prior and concurrent work

We record the overlaps with four items, each examined directly.

Ufnarovski–Åhlander (2003) [7]. Their Theorem 5 gives $p^k \| n$, $0 < k < p \Rightarrow p^{k-1} \| n'$, and their Theorem 4 gives $n^{(j)} \to \infty$ when $p^p \mid n$ properly; together with Corollary 2 (n squarefree $\Leftrightarrow (n, n') = 1$) these force a period–two point to have squarefree members with disjoint supports. The argument is sound: for $2 \leq k < p$ two applications of Theorem 5 give $v_p(n'') = k - 2 \neq k$, while $k \geq p$ falls to Theorem 4 or to the fixed point $n = p^p$, and the edge cases $p = 2$, $k \in \{2, 3\}$ are covered by the latter because $4 = p^p$. They then state the period–two case as **Conjecture 4**: $\left(\sum_{i \leq k} 1/p_i\right)\left(\sum_{j \leq l} 1/q_j\right) = 1$ has no solution in distinct primes. This is #307, and priority for the two–cycle reformulation belongs here rather than to 2019: what [19] adds is the identification of that equation as #307, which [7] does not make — they cite [1] for the derivative, not [2] for the problem. The consequence is that #307 has been an open conjecture in the arithmetic–derivative literature since 2003 under another name, and that the bounds of Section 4 are bounds on Conjecture 4.

Seva (MathOverflow 323194, 2019) [19]. Defines $s(Q) = Q\sum_{p|Q} 1/p$, derives the valuation property $v_p(s(Q)) = v_p(Q) - 1$, asks for $s(P) = Q$, $s(Q) = P$, and states this is Problem #307 "in disguise." Priority for *naming* the two–cycle question as #307 belongs here. He further asks whether every s–trajectory stabilises at 0; the s–analogue of the Ufnarovski–Åhlander conjecture, posed independently of it. (Note $s \neq n'$ globally, e.g. $s(p^\alpha) = p^{\alpha-1}$, but $s = n'$ on squarefree integers, where both two–cycle questions live.)

Kovič (2012), Propositions 17 and 18 [8]. Proposition 18 claims $r + s \geq 34$ unconditionally (≥ 57 if $\min\{p_1, q_1\} \geq 3$, ≥ 110 if ≥ 5), together with $p_1 < r$, $q_1 < s$ and $\max\{m, n\} \geq \prod_{i\leq 17} p_i \approx 1.92 \times 10^{21}$. Its proof does not establish them. Substituting $n' = m$, $m' = n$ into its two correct pairs of bounds gives

$$\text{(A)}\quad \frac{rn}{p_r} < m < \frac{rn}{p_1}, \qquad \text{(B)}\quad \frac{sm}{q_s} < n < \frac{sm}{q_1},$$

after which it asserts "Hence: $p_1q_s < rs$ and $q_1p_r < rs$", and deduces everything from these via $2\max\{p_r, q_s\} < N^2/4$ against $\max\{p_r, q_s\} \geq P_N$. But products of inequalities are valid only in matching directions, and (A) with (B) admits exactly two: upper × upper gives $p_1q_1 < rs$, lower × lower gives $rs < p_rq_s$. The asserted $p_1q_s < rs$ pairs an upper bound on m, carrying p_1, with a lower bound on n, carrying q_s. Nor is it recoverable: the cycle says precisely $\sigma(n) \in (r/p_r,\, r/p_1)$ and $\sigma(m) = 1/\sigma(n) \in (s/q_s,\, s/q_1)$, so $\sigma(n)$ lies in $(r/p_r,\, r/p_1) \cap (q_1/s,\, q_s/s)$; nonemptiness of that intersection is equivalent to those two products and to nothing further, whereas $p_1q_s < rs$ reads $q_s/s < r/p_1$, a comparison of the two *upper* endpoints. In $n = 2\cdot5\cdot13\cdot37\cdot53\cdot73\cdot79\cdot89\cdot97\cdot101$, $m = 3\cdot7\cdot19\cdot31\cdot43\cdot59\cdot67\cdot71\cdot83\cdot103$ — squarefree, disjoint supports, $r = s = 10$ — all four inequalities hold while $p_1q_s = 206 > 100 = rs$ and $q_1p_r = 303 > 100$ (`code/kovic_prop18_audit.gp`; the valid products and the invalidity are machine–checked in `lean/Erdos307/KovicProp18.lean`). Two arithmetic slips sit alongside: Table 2 lists $P_{11} = 27$, and the "product of the first 17 primes" is printed with 27 for 29.

What fails is the *inference*, not the *statement*: no two–cycle is known, so no counterexample to $r+s \geq 34$ exists to be given, and none is claimed. The conclusion is unproven rather than false, and is in fact true, since Theorem 4.3 gives $|U| \geq 60$ by a route touching none of this. Propositions 16 and 19 are sound, and so is 17, but 17 assumes the *smaller* member is odd and [8] notes that no estimate follows when it is even. The strongest valid unconditional bound in the prior literature is therefore that paper's computer search $x > 10^4$; Theorem 4.3 improves it by 52 orders of magnitude, and $|U| \geq 60$ is the first valid cardinality bound for arithmetic–derivative two–cycles rather than an improvement on an earlier one. None of this diminishes [8], which remains a standard reference and whose Propositions 16, 17 and 19 we use.

Bado (preprint, Aix–Marseille, 2026) [26]. Working with $A(S) = \sum_{p\in S} M(S)/p$ and $M(S) = \prod_{p\in S} p$ (our N_S, D_S), Bado independently obtains: the lowest–terms lemma (his Lemma 3.1, our Lemma 2.1); the forcing identities $A(P) = M(Q)$, $A(Q) = M(P)$ with $P\cap Q = \varnothing$ (his Thm. A, our Theorem 2.3); the two–cycle reformulation $T(T(P)) = P$ (his Thm. B); the valuation signature (his Thm. 6.1, = Israel's); the local zero–sum congruences (his Prop. 8.1, our Lemma 2.1); the parity restrictions (his Prop. 8.2, the parity part of Theorem 4.18); an exact one–sided test (his Thm. 9.1); the deviation parameter, i.e. the integer t with $A(P) = n + tm$, $A(Q) = m - tn$ and solubility exactly at $t = 0$ (his Thm. 7.6, our rung k of Proposition 5.1); the intrinsic union criterion (his Thm. 7.3, our Proposition 10.1 in *iff* form); uniqueness of the splitting (his Cor. 7.5); the approximation theorem (his Thm. 10.2, sharper than our greedy version in that all primes exceed a given bound), and the union discriminant / Pythagorean structure (his Thms. 7.1, 7.4). He cites only

Barbeau, Bloom, and Erdős–Graham: crucially, he does *not* identify the map with the arithmetic derivative, and his note contains none of the present note's $n'' = n$ bridge, explicit barrier constant, function–field theorem, density theorem, k–cycle bounds, or constructive deficit analysis. We regard the overlap on the shared structural layer as strong independent validation, and import his union criterion with credit. Bado has confirmed this account in correspondence (August 2026): he agrees that the results listed above were present in his preprint and that the overlap is concurrent and independent, and he has confirmed the identification $A(S) = M(S)'$, which he had not made, and is revising his historical discussion accordingly. The concurrency is therefore recorded on both sides rather than asserted on one:

Proposition 10.1 (Single–integer test; derivative form of Bado's Thms. 7.1, 7.4)**.** *If (a, b) is a two–cycle of the arithmetic derivative, then $N = ab$ is squarefree and satisfies $N' = a^2 + b^2$, whence $N'^2 - 4N^2 = (a^2 - b^2)^2$ is a perfect square. Conversely, a squarefree N with $N'^2 - 4N^2 = \square$ determines a unique unordered factorisation $N = ab$ with $N' = a^2 + b^2$, on which the cycle condition reduces to a single equation (Bado's deviation parameter $t = 0$, his Thm. 7.6). This gives a one–integer necessary test; computationally, no squarefree $N < 2 \times 10^5$ with $\omega(N) \geq 2$ passes it (*0 *of* 103,596*, consistent with Theorem 4.3, which forces $N > 4 \times 10^{112}$).*

Proof. $N = ab$ with a, b coprime squarefree gives N squarefree and, by Leibniz with $a' = b$, $b' = a$, $N' = a'b + ab' = b^2 + a^2$; then $N'^2 - 4N^2 = (a^2 + b^2)^2 - 4(ab)^2 = (a^2 - b^2)^2$. The converse is Bado's [26]; the emptiness of the test below 10^{112} is upgraded from a sieve to a theorem in Corollary 10.3. □

The Pythagorean equation $N' = a^2 + b^2$ deserves to be isolated from the (harder) cycle condition.

Proposition 10.2 (Pair form; the one–equation relaxation)**.** *For coprime squarefree a, b, the single equation $a^2 + b^2 = (ab)'$ holds if and only if there is an integer k with $a' = b + ak$ and $b' = a - bk$; a two–cycle is exactly the case $k = 0$. We call a solution of $a^2 + b^2 = (ab)'$ (any k) a* Pythagorean pair*.*

Proof. The Leibniz identity $a^2 + b^2 - (ab)' = a(a - b') - b(a' - b)$ (a polynomial identity, verified symbolically and on 2000 random pairs) makes the equation equivalent to $a(a-b') = b(a'-b)$; since $\gcd(a, b) = 1$ this forces $a \mid a' - b$ and $b \mid a - b'$ with equal quotients k, i.e. $a' = b + ak$, $b' = a - bk$. Setting $k = 0$ recovers $a' = b$, $b' = a$. (Equivalently, in $\mathbb{Z}[i]$: #307 asks for $z = a + bi$ with coprime coordinates and ab squarefree satisfying $|z|^2 = (\operatorname{Im} z^2/2)'$ and $k = 0$.) □

Corollary 10.3 (The single–integer test is provably empty below 10^{112})**.** *Every Pythagorean pair satisfies $\sum_{p|a} 1/p + \sum_{q|b} 1/q = a/b + b/a \geq 2$ (divide $a^2 + b^2 = (ab)'$ by ab and apply AM–GM), so its support carries at least* 59 *primes and $N = ab \geq \prod($first 59 primes$) > 7.9 \times 10^{112}$. Hence no squarefree $N < 10^{112}$ with $\omega(N) \geq 2$ satisfies $N'^2 - 4N^2 = \square$: the empirical "*0 *of* 103,596*" of Proposition 10.1 is a theorem, and the barrier of Theorem 4.3 extends from two–cycles to the strictly weaker one–equation (Pythagorean) problem.*

Formalised. *`Erdos307.pyth_sum_of_squares` is $N' = a^2 + b^2$ from Leibniz and the cycle, `pyth_discriminant` the resulting square, `split_identity` and `split_product` the two–layer form of Proposition 5.13, and `mass_ge_two_of_pythagorean` with `card_ge_59_of_pythagorean` this corollary, glued by `mass_ge_two_of_pyth_support` into `emptytest`, which takes the Pythagorean equation itself rather than an assumed mass bound. Earlier editions proved the two ends and not the link, so this tag overstated what was present. The AM–GM step of Proposition 4.19 is `sum_ge_card_of_prod_eq_one`, valid for every k at once and proved from $\log x \leq x - 1$ rather than a k–term AM–GM. Bado's converse is cited, not reproved* (`lean/Erdos307/Pythagorean.lean`).

Remark 10.4. (a) As $N' = a^2 + b^2$ is a sum of two coprime squares, every odd prime dividing N' is $\equiv 1 \pmod 4$; a cheap union–side pre–filter complementing Bado's congruence filters. (When exactly one of a, b is even this is the hypotenuse of the primitive triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$.) (b) Since Seva's s agrees with n' on squarefree integers, and his valuation property forces two–cycle members squarefree, his question is *exactly* #307, not merely an analogue. (c) Primitive Pythagorean triples form the Barning–Hall ternary tree (the orbit structure of a reflection group in $O(2, 1; \mathbb{Z})$), and a #307 solution's triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$ occupies one node with generator $(a, b) = (\prod P, \prod Q)$. The tree nonetheless yields no descent on the problem: its moves $(a, b) \mapsto (2a \mp b,\, a)$, $(a + 2b,\, b)$ are linear, preserving coprimality but not squarefreeness (over random squarefree coprime generators only 37% of children keep both members squarefree) so the locus of #307–eligible nodes (squarefree generators of disjoint prime support meeting $a^2 + b^2 = (ab)'$) is not tree–invariant. The additive tree is transverse to the multiplicative constraint; the arithmetically natural recursion is instead the prime–append δ–tree of Proposition 7.33, $\delta(Mp) = p\,\delta(M) + M$, which preserves squarefreeness and tracks the line parameter, which is why the classical members organise on *that* tree, not Barning–Hall.

The deviation parameter of Proposition 10.2 organises the one–equation problem into a short graded family.

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